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1779782466869_2605_03860v1_pdf

SOURCE FILENAME
2605.03860v1.pdf

VERSION TAG
v1.0

Benchmark overview

Scores indicate framework-relative alignment only. They do not represent objective moral truth or a universal ethical score.

STRONGEST ALIGNMENT
Utilitarianism, Contractualism

WEAKEST ALIGNMENT
Deontology

Cross-framework pattern. Across frameworks, the use case is widely credited for replacing ad hoc curtailment with a transparent and principled fairness framework. The main divide is between frameworks that accept welfare-based optimization with explicit fairness structure and frameworks that insist that rights, duties, participation, or vulnerability cannot be reduced to optimization inputs. Similar mid-range scores often reflect different concerns: some focus on legitimacy and participation, others on care, character, or community meaning.

Overall ethical position. The use case occupies a comparatively strong but incomplete position in the ethical landscape. It is a serious and defensible fairness framework for constrained grid allocation, but it requires stronger rights protection, participatory justification, and attention to vulnerability to achieve broader ethical alignment.

Deontology — Serious Tension

Alignment: 35 **Concern:** 70 **Confidence:** 85

Utilitarianism — Moderate To Strong Alignment, But With Notable Unresolved Concerns

Alignment: 75 **Concern:** 45 **Confidence:** 80

Rawlsian Justice — Mixed

Alignment: 65 **Concern:** 45 **Confidence:** 80

Rights-Based Ethics — Mixed

Alignment: 55 **Concern:** 65 **Confidence:** 70

Contractualism — Moderate Alignment

Alignment: 70 **Concern:** 60 **Confidence:** 75

Principlism Mixed

Alignment: 65 Concern: 40 Confidence: 70

Virtue Ethics Mixed

Alignment: 55 Concern: 60 Confidence: 70

Care Ethics Mixed

Alignment: 55 Concern: 60 Confidence: 70

Communitarian Ethics Moderate Alignment

Alignment: 65 Concern: 45 Confidence: 70

Casuistry Moderate Alignment

Alignment: 65 Concern: 45 Confidence: 70

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Summary

Ethical Landscape Synthesis for: A Welfarist Perspective on Fair Generation Curtailment

1) Introduction

This use case proposes an axiomatic method for deciding how to curtail distributed solar generation when local grid constraints make full export impossible. Instead of relying on ad hoc fairness metrics, it models curtailment as a social choice problem and argues that, under a cardinal non-comparability stance on utilities, the Kalai-Smorodinsky (KS) rule provides a principled way to allocate limited grid access. The ethical ambition is substantial: to make curtailment not only technically feasible, but also fair, transparent, auditable, and investment-compatible. Across the evaluation reports, the central question is not whether fairness matters, but what kind of fairness this framework actually delivers, what it leaves out, and whether its welfarist structure can bear the moral weight it claims.

2) Convergences

A strong point of convergence is that many frameworks regard the use case as ethically better than opaque or ad hoc curtailment practices because it makes normative choices explicit. This is one of the clearest cross-framework strengths. The framework identifies fallback and utopia points, ties them to recognizable claims such as self-consumption or export entitlements, and shows how different curtailment rules can be understood as principled variants rather than arbitrary heuristics. That transparency is praised by deontological, Rawlsian, utilitarian, and communitarian evaluations alike, though for different reasons: as moral clarity [Deontology eval], publicity and public justification [Rawlsian Justice eval], rational welfare design [Utilitarianism eval], and improved legitimacy within shared institutions [Communitarian Ethics eval].

A second convergence is that the use case is widely seen as ethically serious in its attempt to avoid crude aggregation across persons. The decision to reject full interpersonal comparability of utilities is especially important here. Several reports treat this as a meaningful moral improvement over standard welfare maximization. Deontology sees in it at least a partial respect for the separateness of persons, even if not enough to satisfy rights-based constraints [Deontology eval]. Rawlsian Justice reads it as sensitivity to pluralism and heterogeneous circumstances [Rawlsian Justice eval]. Contractualism sees it as closer to taking individual complaints seriously rather than simply summing them [Contractualism eval]. Even the utilitarian evaluation, though more favorable overall, treats this move as a pragmatic and morally relevant response to the epistemic weakness of comparing household welfare on a single scale [Utilitarianism eval].

A third area of agreement is that the framework captures an important fairness intuition: when a shared public resource is scarce, burdens should not be imposed in an arbitrary or systematically biased way. The KS rule's focus on equalizing relative gains from a baseline is interpreted by several frameworks as a plausible way of expressing non-discrimination and reciprocity in access to the grid [Rawlsian Justice eval; Principlism eval; Contractualism eval]. The use case is therefore seen as ethically stronger than purely utilitarian curtailment rules that minimize total curtailment while allowing highly unequal burdens based on network position [Utilitarianism eval; Principlism eval].

3) Key Ethical Tensions

The deepest fault line concerns whether fairness can be adequately represented through a welfarist optimization rule at all. Utilitarianism is broadly sympathetic because the framework is explicitly consequentialist and welfare-oriented, even if it departs from classical utilitarianism by weakening interpersonal comparability and prioritizing relative gains over total welfare [Utilitarianism eval]. By contrast, deontology and rights-based ethics argue that this is precisely the problem: the framework treats moral claims as inputs to an optimization procedure rather than as constraints that may not be overridden [Deontology eval; Rights-Based Ethics eval]. On this view, a fallback point is not yet a right unless it is protected categorically.

A second fault line concerns justice as distribution versus justice as legitimacy. Rawlsian, communitarian, care-ethical, and contractualist evaluations all note that the framework is strongest on distributive structure and weakest on participatory justification. It says a great deal about how to rank feasible allocations, but much less about who gets to define the utility proxy, the fallback entitlement, or the utopia benchmark [Rawlsian Justice eval; Care Ethics eval; Communitarian Ethics eval; Contractualism eval]. Similar scores across these frameworks conceal different concerns: Rawlsian justice worries about public reason and fair equality of opportunity; communitarian ethics worries about community values and shared meaning; care ethics worries about responsiveness to vulnerable persons and lived dependency; contractualism worries about whether affected parties could reasonably reject the governing principle.

A third tension lies between fairness and welfare efficiency. The use case criticizes fairness metrics that violate Weak Pareto and thereby create unnecessary curtailment. Yet the utilitarian evaluation still notes that the KS rule may sacrifice aggregate welfare relative to total-welfare-maximizing alternatives, because maximizing the minimum relative gain is not the same as maximizing total benefit [Utilitarianism eval]. This is a classic trade-off between equality-sensitive allocation and aggregate efficiency. The framework handles one inefficiency problem well—Pareto-dominated fairness rules—but does not eliminate the broader moral tension between equal treatment and total output.

4) Robust Improvements

The most robust improvement across frameworks would be to convert selected fallback benchmarks into explicit side-constraints rather than merely reference points in optimization. If self-consumption, minimum export, or some baseline access is treated as a protected floor that cannot be violated except under clearly specified emergency conditions, alignment would improve for deontology, rights-based ethics, principlism, and likely contractualism [Deontology eval; Rights-Based Ethics eval; Principlism eval; Contractualism eval]. The trade-off is reduced flexibility and, in some cases, lower aggregate efficiency or feasibility margins.

A second broadly beneficial revision would be to institutionalize participatory selection of utility proxies and reference points. The reports repeatedly identify these design choices as ethically decisive and currently under-justified. A deliberative process involving prosumers, operators, and regulators would improve Rawlsian legitimacy, communitarian fit, care-ethical responsiveness, and contractualist justifiability [Rawlsian Justice eval; Communitarian Ethics eval; Care Ethics eval; Contractualism eval]. It would also help utilitarianism indirectly by improving the empirical validity of welfare proxies [Utilitarianism eval]. The trade-off is procedural complexity and slower deployment.

A third robust improvement would be to incorporate vulnerability-sensitive and longitudinal features. Several reports note that static, snapshot fairness may miss cumulative burdens, socio-economic disadvantage, or crisis conditions [Care Ethics eval; Rawlsian Justice eval; Principlism eval; Utilitarianism eval]. Tracking curtailment over time, identifying especially dependent or disadvantaged prosumers, and adding emergency-response rules would improve alignment across care ethics, Rawlsian justice, principlism, and virtue ethics. The trade-off is that stronger responsiveness to need may weaken strict anonymity or simple equal-relative-gain rules.

5) Where Ethical Alignment May Not Be Possible

Some tensions are not fully reconcilable. The clearest is between optimization-based fairness and rights as side-constraints. If the framework is made strongly rights-protective, it will become less able to trade off burdens for system-wide welfare or equal-relative-gain fairness; if it remains a welfare-ranking system, deontological and rights-based objections will persist [Deontology eval; Rights-Based Ethics eval]. Decision-makers must choose whether the grid is governed primarily by protected entitlements or by a fairness-sensitive optimization of feasible outcomes.

A second non-reconcilable tension concerns impartial formal fairness versus context-sensitive responsiveness. KS-style equalization of relative gains offers a clean, auditable rule. But care ethics, virtue ethics, and communitarian ethics all suggest that morally relevant differences in dependence, community meaning, or vulnerability may justify departures from uniform treatment [Care Ethics eval; Virtue Ethics eval; Communitarian Ethics eval]. A system can be more context-sensitive or more formally uniform, but not maximally both.

6) Framework-relative alignment

Taken together, the ethical landscape is neither a rejection nor an endorsement. The use case is strongest where fairness requires transparent, non-arbitrary, and non-crudely aggregative allocation under technical scarcity. It is weakest where ethics requires rights protection, participatory legitimacy, or richer attention to vulnerability, agency, and moral character. The broad pattern is that frameworks friendly to structured welfare reasoning view the proposal as promising but incomplete, while frameworks centered on duties, rights, or relationships see serious but potentially improvable limitations.

Evaluations

Deontology

Framework evaluation

ALIGNMENT	CONCERN	CONFIDENCE	BENCHMARK
35	70	85	**Serious Tension** The use-case's consequentialist welfare maximization and utility aggregation fundamentally conflict with deontological demands for categorical duties and respect for agency and rights. While it partially respects individual baseline entitlements, it lacks mechanisms to uphold inviolable moral norms, leading to serious ethical tensions under deontology.

1) The Use-Case, Reconstructed

The use-case under examination is a proposed welfarist approach to fair active power curtailment in distribution grids with distributed photovoltaic (PV) generation, as presented in the PowerUp Conference 2026 paper titled "A Welfarist Perspective on Fair Generation Curtailment" by Jonas G. Matt, Ilia Shilov, and Saverio Bolognani. The decision problem addressed is the allocation of limited grid capacity among multiple prosumers (consumers who also produce energy) when the total available generation exceeds the grid's operational constraints, necessitating active power curtailment.

The mechanism proposed is a principled, axiomatic framework based on social choice theory that models curtailment decisions as a social choice problem over feasible operating points. The key innovation is the derivation of curtailment objectives from foundational axioms expressing fairness and grid access rights, rather than relying on ad-hoc or heuristic fairness metrics. The framework adopts a cardinal non-comparability stance on utilities, meaning that while utilities are cardinally measurable for each agent, they are not assumed to be interpersonally comparable in levels or increments. This stance avoids problematic assumptions about the commensurability of heterogeneous prosumers' utilities.

The core solution is the use of the Kalai-Smorodinsky (KS) social welfare function to select curtailment profiles. The KS rule maximizes the minimum relative gain of each prosumer from a defined fallback utility (a baseline, e.g., zero generation or self-consumption) toward a utopia utility (the ideal maximum generation if unconstrained). This approach ensures fairness by equalizing the relative improvement each agent obtains, respecting individual entitlements and grid constraints. The framework unifies several existing curtailment schemes as special cases of the KS rule with different fallback and utopia points, thus providing a rigorous axiomatic foundation for these methods.

The target outcomes are multiple: to ensure fairness in grid access, to respect sustainability goals by minimizing unnecessary curtailment, to promote social equity by sharing the public resource of the grid fairly, and to encourage investments by providing clear, auditable, and principled curtailment rules. Success is measured by achieving Pareto efficient curtailment decisions that are transparent, justifiable, and aligned with regulatory and social objectives.

Implicit value claims include the prioritization of fairness understood as equal relative gains from baseline entitlements, respect for individual rights to self-consumption or export, and the avoidance of arbitrary or opaque decision-making. The approach also implicitly values transparency and auditability, enabling grid operators and stakeholders to understand and justify curtailment decisions.

The decision context involves multiple stakeholders: prosumers with heterogeneous generation capacities and demands, grid operators responsible for maintaining operational constraints, regulators mandating non-discriminatory access, and society at large concerned with sustainability and equity. The optimization privileges fairness as defined by the KS social welfare function, balancing individual entitlements and collective feasibility.

The use-case assumes the availability of utility proxy metrics (e.g., generated power or export) for each prosumer, a convex and monotonic feasible set of operating points defined by grid constraints, and the ability to identify fallback and utopia envelopes. It also assumes that utilities are cardinally measurable for each agent but not interpersonally comparable, which is critical for the choice of the KS rule.

In summary, the use-case offers a theoretically grounded, axiomatic, and computationally tractable framework for fair PV curtailment in distribution grids, aiming to reconcile competing criteria of fairness, efficiency, and investment promotion under realistic assumptions about utility comparability.

2) The Framework, Reconstructed

The ethical framework to be applied in evaluating this use-case is Deontology, a normative moral theory grounded in the concept of duty and the categorical nature of moral norms. Deontology derives from the Greek words for duty (deon) and study (logos), and it is centrally concerned with which choices are morally required, forbidden, or permitted, independent of their consequences. The framework contrasts with consequentialism, which assesses the morality of actions solely by their outcomes.

At its core, deontology holds that certain actions are categorically impermissible regardless of the good consequences they might produce. Moral agents have duties to act in accordance with moral norms, which have priority over considerations of the good (the Right has priority over the Good). These duties are not to be maximized or traded off against one another in a consequentialist calculus but are to be obeyed categorically.

Deontological theories are diverse but can be broadly categorized into agent-centered and patient-centered theories. Agent-centered theories emphasize the moral significance of the agent's own agency, intentions, and actions, often focusing on the agent-relative reasons and permissions that allow for partiality toward one's own projects and relationships. Patient-centered theories focus on the rights of individuals (patients) and prohibit using others as mere means without consent, emphasizing respect for persons' bodily and labor rights.

The framework requires that moral norms be obeyed categorically, with no exceptions justified by appealing to better overall consequences. It also demands that moral agents respect the separateness of persons, prohibiting the aggregation of harms or rights violations across individuals to justify wrongdoing. Deontology allows for permissions and supererogatory acts but insists on strict boundaries around forbidden acts, such as killing innocents or violating rights.

Deontology faces challenges such as the paradox of deontology (how to handle situations where violating a duty prevents more violations), conflicts between duties, and moral catastrophes where adherence to duty leads to disastrous consequences. Responses include specification of duties to avoid conflicts, threshold deontology allowing exceptions beyond certain limits, and the doctrine of double effect distinguishing intended from foreseen harms.

The framework treats duties, rights, and permissions as morally primary, focusing on the agent's conformity to moral norms rather than on the maximization of welfare or utility. It demands clarity about the content and scope of duties and the conditions under which they apply, emphasizing the moral significance of agency, intention, and respect for persons.

Typical failure modes in deontology include ignoring the separateness of persons by aggregating harms, failing to respect agent-relative reasons and permissions, allowing consequentialist reasoning to override categorical duties, and permitting conflicts of duties without resolution. The framework requires rigorous justification for any exceptions to duties and demands that moral agents maintain the integrity of their own agency and respect the rights of others.

In sum, deontology is a normative ethical theory that prioritizes categorical duties and rights over consequentialist calculations, emphasizing respect for persons, the moral significance of agency and intention, and the inviolability of certain moral norms.

3) Principle-by-Principle Deep Critical Evaluation

The evaluation of the welfarist approach to fair generation curtailment through the lens of deontology requires a careful examination of how the use-case aligns with or diverges from the core commitments of deontological ethics. This involves analyzing the use-case's treatment of duties, rights, permissions, agency, and the moral significance of actions independent of consequences.

The first principle of deontology relevant here is the categorical nature of moral duties. Deontology demands that certain actions be categorically forbidden or required, regardless of their outcomes. The use-case, however, is explicitly welfarist and consequentialist in orientation: it models curtailment decisions as social choices aimed at maximizing a social welfare function (the Kalai-Smorodinsky rule) over feasible operating points. This approach inherently aggregates utilities and seeks Pareto efficient outcomes, privileging states of affairs that maximize relative gains. Thus, the use-case fundamentally evaluates the morality of curtailment decisions by their consequences for prosumers' utilities, rather than by adherence to categorical norms.

From a deontological standpoint, this is a significant divergence. Deontology requires that the morality of an act be assessed by its conformity to moral norms, not by the aggregate welfare it produces. The use-case's reliance on social welfare functions and Pareto efficiency aligns with consequentialist rationality, which deontology explicitly rejects as the primary moral standard. Therefore, the use-case does not satisfy the deontological requirement that the Right have priority over the Good; instead, it privileges the Good (maximizing welfare) over categorical duties.

Second, deontology emphasizes the moral significance of agency and intention. Agent-centered deontological theories hold that agents have agent-relative obligations and permissions grounded in their own agency, intentions, and actions. The use-case, however, abstracts away from individual agency and intention, focusing instead on utility profiles and social welfare rankings. It does not address whether curtailment decisions respect the agents' moral agency or whether the agents consent to the use of their generation capacity in particular ways. The framework assumes that utilities can be measured or proxied but does not consider whether the agents' rights or intentions are respected in the curtailment process.

This absence of attention to agency and intention is a further departure from deontological ethics, which requires that agents not be treated merely as means to an end without their consent. The use-case's focus on utility maximization risks instrumentalizing prosumers as mere contributors to aggregate welfare, potentially violating their rights or agency if their generation is curtailed without respecting their moral claims.

Third, patient-centered deontology, with its emphasis on rights against being used without consent, raises questions about whether the use-case respects prosumers' rights to their generation capacity. The use-case does include normative reference points such as fallback envelopes that can be interpreted as entitlements (e.g., self-consumption rights). This suggests some recognition of individual rights or baseline claims. However, these rights are embedded within a consequentialist framework that allows for their violation if doing so improves overall welfare or fairness as measured by the KS rule.

Deontology would require that such rights be inviolable or only overridden under strictly defined and justified conditions, not merely balanced against welfare gains. The use-case does not articulate categorical prohibitions against violating prosumers' rights, nor does it provide a mechanism to respect these rights independently of utility considerations. This indicates a misalignment with the deontological insistence on the priority and inviolability of rights.

Fourth, the use-case's adoption of cardinal non-comparability of utilities aligns with a deontological sensitivity to the separateness of persons. Deontology rejects the aggregation of harms or benefits across individuals as a justification for wrongdoing. By refusing to assume interpersonal comparability of utilities, the use-case avoids some of the problematic aggregations that consequentialism permits. This is a point of partial alignment: the use-case respects the idea that utilities are agent-relative and not directly commensurable, which resonates with deontology's concern for respecting individual moral standing.

However, the use-case still aggregates utilities through the KS social welfare function, which equalizes relative gains and thus implicitly compares individuals' welfare improvements. While it does not assume full comparability, it does assume enough comparability to rank social states and make trade-offs. Deontology would question whether such trade-offs are permissible when they involve violating categorical duties or rights. Thus, the use-case's stance on comparability is a nuanced partial alignment but does not fully satisfy deontological constraints against aggregating across persons in morally significant ways.

Fifth, the use-case's reliance on fallback and utopia envelopes as normative reference points can be interpreted as attempts to ground curtailment decisions in baseline entitlements and ideal aspirations. This is a promising feature from a deontological perspective, as it acknowledges that agents have claims that must be respected as starting points. The fallback envelope can be seen as a minimal right or duty baseline, while the utopia envelope represents what agents could claim in the absence of constraints.

Nonetheless, the use-case treats these reference points as parameters in a welfare maximization problem, not as categorical constraints or rights that must be respected regardless of consequences. Deontology requires that such rights or duties be upheld categorically, not overridden by welfare considerations. The use-case's framework allows for curtailment decisions that may reduce some agents' utilities below their fallback levels if overall welfare or fairness demands it, which would be impermissible under deontological norms.

Sixth, deontology is concerned with conflicts of duties and the possibility of moral dilemmas. The use-case does not explicitly address conflicts of moral duties or categorical prohibitions; instead, it resolves allocation problems through optimization and social welfare maximization. This approach sidesteps the issue of conflicting duties by subsuming all considerations under a welfare function, which deontology would find problematic. The absence of a mechanism to handle duty conflicts or to respect categorical prohibitions indicates a divergence from deontological methodology.

Seventh, the use-case's emphasis on transparency, auditability, and principled design choices aligns with deontology's concern for moral clarity and the rule-governed nature of morality. Deontology values clear, consistent norms that agents can follow and that respect the integrity of moral agency. The use-case's axiomatic foundation and explicit parameterization of fallback and utopia points provide a transparent basis for decision-making, which is commendable from a deontological perspective.

However, transparency and principled design do not compensate for the fundamental consequentialist orientation of the framework. Deontology requires that moral norms be categorical and not subject to optimization or balancing against welfare. Thus, while the use-case's transparency is a strength, it does not resolve the deeper misalignment with deontological principles.

Eighth, the use-case's treatment of fairness as equalizing relative gains resonates with deontological concerns for equality and respect for persons. Deontology often emphasizes the equal moral worth of individuals and the importance of treating persons as ends in themselves. The KS rule's focus on maximizing the minimum relative gain can be seen as an attempt to respect this equality.

Yet, deontology would insist that fairness also entails respecting inviolable rights and duties, not merely equalizing welfare outcomes. The use-case's fairness conception is consequentialist in nature, aiming at equitable welfare distribution rather than categorical respect for rights. This represents a partial alignment in valuing fairness but a divergence in the conception and grounding of fairness.

Finally, the use-case does not address the metaethical commitments of deontology, such as the priority of norm conformity over welfare maximization, the moral significance of intentions, or the separateness of persons as moral agents. It also does not engage with the deontological challenges of moral catastrophes or threshold deontology, which would be relevant in grid management scenarios involving trade-offs between harms.

In sum, the use-case aligns with deontology in its recognition of individual baseline entitlements, its sensitivity to interpersonal comparability, and its commitment to transparency and fairness. However, it diverges fundamentally in its consequentialist orientation, its aggregation of utilities, its lack of categorical duties or rights, and its neglect of agency and intention. These divergences are intrinsic to the use-case's welfarist framework and cannot be fully reconciled with deontological ethics without structural changes.

4) Stress Tests / Edge Cases

To further probe the alignment of the use-case with deontological ethics, consider two stress test scenarios that highlight potential ethical tensions.

The first scenario involves a prosumer whose fallback utility corresponds to self-consumption rights, ensuring a minimal entitlement to use their generated power for their own demand. Suppose that, due to grid constraints, the curtailment scheme based on the KS rule requires this prosumer's generation to be curtailed below their fallback level to maximize the minimum relative gain across all prosumers. This would mean violating the prosumer's baseline entitlement.

From a deontological perspective, this is impermissible. The fallback envelope represents a categorical right or duty baseline that must not be overridden by welfare considerations. The use-case's willingness to allow curtailment below fallback levels in pursuit of social welfare maximization conflicts with the deontological requirement to respect categorical duties and rights. This reveals a principled flaw: the use-case's framework lacks mechanisms to enforce inviolable rights or duties, risking morally impermissible outcomes.

The second scenario concerns a prosumer who invests in upgrading their generation capacity, thereby improving their utopia utility. According to the Individual Monotonicity axiom, the KS solution should not reduce this prosumer's utility following such an improvement. However, consider a situation where the grid constraints are so tight that accommodating the improved capacity of this prosumer requires more severe curtailment of others, reducing their relative gains significantly.

While the KS rule respects Individual Monotonicity for the improving prosumer, it does so at the expense of others' utilities. From a deontological standpoint, this raises concerns about fairness and respect for all persons' rights. The framework's focus on relative gains and welfare maximization may permit trade-offs that violate categorical duties to others, such as not to harm or use them as mere means.

This scenario exposes the tension between agent-relative duties and agent-neutral welfare aggregation. Deontology would require that the rights and duties owed to each prosumer be respected independently, not overridden by aggregate welfare gains. The use-case's framework, by contrast, permits such trade-offs, revealing an intrinsic misalignment with deontological ethics.

These stress tests illustrate that the use-case's consequentialist aggregation and welfare maximization approach can lead to violations of categorical rights and duties, which deontology forbids. The framework lacks safeguards to uphold inviolable moral norms, indicating that minor amendments would be insufficient to resolve these tensions; structural changes would be necessary.

5) Recommendations

Given the fundamental divergences identified between the use-case's welfarist framework and deontological ethics, recommendations focus on how to improve alignment or clarify the limits of the use-case's moral claims.

First, to enhance alignment, the use-case could incorporate explicit categorical constraints representing inviolable rights or duties. For example, the fallback envelope could be elevated from a normative parameter to a categorical floor that must not be violated under any circumstances. This would reflect deontological respect for baseline entitlements and prevent curtailment decisions that infringe on prosumers' rights to self-consumption or minimal generation.

Second, the framework should explicitly recognize and incorporate agent-relative permissions and obligations, acknowledging the moral significance of individual agency and intentions. This might involve mechanisms to ensure that curtailment decisions respect prosumers' moral agency, such as requiring consent or providing compensatory permissions when curtailment is necessary.

Third, the use-case should address the possibility of conflicts between duties, providing a principled method for resolving such conflicts without resorting solely to welfare maximization. This could involve specification of duties with priority rules or adopting a threshold deontology approach where categorical duties hold except in extreme cases.

Fourth, the framework should clarify the moral status of the social welfare function and the KS rule, acknowledging that these represent a consequentialist approach that may conflict with deontological norms. Transparency about these limitations would prevent overclaiming the moral authority of the framework from a deontological perspective.

Fifth, the use-case could benefit from integrating a rights-based perspective alongside the welfarist approach, recognizing that fairness involves respecting individual rights as categorical constraints rather than merely balancing utilities. This hybrid approach could better capture the moral complexity of grid curtailment decisions.

Finally, if structural changes to incorporate deontological principles are infeasible, the use-case should explicitly acknowledge its consequentialist foundation and limit its moral claims accordingly. This would provide clarity about the ethical assumptions underlying the framework and avoid conflating consequentialist welfare maximization with deontological duty-based ethics.

In conclusion, while the use-case offers a rigorous and transparent framework for fair curtailment decisions, its fundamental consequentialist orientation diverges from deontological ethics. Addressing this divergence requires structural incorporation of categorical duties, rights, and agent-relative considerations, or clear acknowledgment of the framework's ethical scope and limitations.

Utilitariaism

Framework evaluation

ALIGNMENT	CONCERN	CONFIDENCE	BENCHMARK
75	45	80	**Moderate To Strong Alignment, But With Notable Unresolved Concerns** The use-case aligns well with utilitarianism's welfare maximization, impartiality, and consequentialism, innovatively addressing interpersonal comparability via the Kalai-Smorodinsky social welfare function. However, it diverges from classical utilitarian assumptions on full comparability and maximization, relies on normative reference points, and under-specifies welfare measurement, generating meaningful concerns.

1) The Use-Case, Reconstructed

The use-case under evaluation is a proposed welfarist approach to fair active power curtailment in distribution grids with distributed photovoltaics (PV). The core decision problem arises from the increasing penetration of distributed PV generation in local energy systems, which necessitates active power curtailment to respect grid operational constraints such as voltage limits, line current capacities, and transformer ratings. Curtailment decisions must be made by grid operators to determine which generators to curtail and by how much, balancing multiple, often conflicting criteria: sustainability (minimizing total curtailment of renewables), fair grid access (non-discriminatory network access mandated by regulation), social equity (fair sharing of a public resource), and investment promotion (ensuring profitability to encourage renewable investments).

The proposed solution models the curtailment decision as a social choice problem over the set of feasible operating points, where each feasible operating point corresponds to a vector of power generation envelopes for prosumers (producers-consumers). The approach adopts a welfarist perspective, wherein the utilities of individual prosumers—proxied by measurable metrics such as power generation or export—are aggregated into a social welfare function (SWF) to rank feasible operating points. The SWF is derived axiomatically, relying on foundational principles such as Weak Pareto efficiency, Independence of Irrelevant Alternatives (IIA), Continuity, and Anonymity, as well as assumptions about interpersonal comparability of utilities.

Crucially, the use-case rejects the assumption of full interpersonal comparability of utilities, which is often untenable in heterogeneous residential systems. Instead, it adopts a cardinal non-comparability stance (CNC), where utilities are cardinally meaningful within agents but not comparable across agents. This stance requires fewer assumptions about prosumers' private preferences and avoids problematic interpersonal utility comparisons. Under CNC, classical impossibility results preclude a globally defined SWF satisfying all axioms without additional structure. The use-case addresses this by introducing two normative reference points: a fallback envelope (a worst-case benchmark, e.g., zero generation) and a utopia envelope (coordinate-wise maximal utilities). These references enable the definition of the Kalai-Smorodinsky (KS) social welfare function, which maximizes the minimum relative gain of each agent from fallback to utopia, ensuring fairness and efficiency.

The use-case further demonstrates that existing curtailment schemes in the literature can be unified as specific instances of the KS rule with different choices of fallback and utopia points. This unification provides grid operators with an auditable, axiomatic foundation to justify fairness in local energy systems, moving away from ad-hoc fairness indices that may lead to inefficiency or inequity.

The target outcomes are thus: (1) a principled, axiomatic foundation for fair curtailment decisions; (2) a social welfare function that respects fairness, efficiency, and grid access rights without requiring full interpersonal utility comparability; (3) transparency and auditability in curtailment policy design; and (4) practical implementability, demonstrated through simulations on a real-world low-voltage power grid testbed.

Implicit value claims include the primacy of welfare (utility) as the moral metric, the importance of fairness understood as equitable relative gains from baseline to utopia, the rejection of arbitrary or opaque fairness metrics, and the necessity of respecting grid access rights as a form of social equity. The use-case assumes that utilities can be meaningfully proxied by power generation or export metrics and that the feasible set of operating points is convex and monotonic, allowing for free curtailment down to fallback levels.

The stakeholders include prosumers (residential PV owners), grid operators, regulators, and society at large, with competing interests in maximizing renewable energy utilization, ensuring fair access, and maintaining grid stability. The use-case privileges a social welfare maximization approach that balances these interests through an axiomatic, welfarist lens.

Success is measured by achieving feasible curtailment decisions that are Pareto efficient, fair according to the KS criterion, and aligned with regulatory and social objectives, while avoiding inefficiencies and disincentives inherent in prior ad-hoc schemes.

2) The Framework, Reconstructed

The ethical framework to be applied in this evaluation is Utilitarianism, specifically classical utilitarianism as developed by Jeremy Bentham and John Stuart Mill, and its subsequent elaborations including act and rule utilitarianism. Utilitarianism is a form of consequentialism, holding that the moral status of actions depends solely on their consequences, specifically on the extent to which they promote utility—understood classically as happiness or pleasure, or more generally as well-being.

Core principles of utilitarianism include consequentialism, hedonism or welfarism (pleasure or well-being as the sole intrinsic good), aggregation (the sum total of individual utilities determines overall utility), maximization (one ought to choose the action that produces the greatest total utility), impartiality (each individual's utility counts equally), and inclusivity (the welfare of all sentient beings is morally relevant).

The framework treats welfare—understood broadly as well-being or happiness—as the morally primary value. Moral rightness is determined by the maximization of this welfare. The framework demands that individual interests be aggregated impartially, without privileging oneself or particular groups, and that the best action is that which maximizes total welfare.

Scope conditions include the applicability of utilitarianism to individual and collective moral decisions, including social policies and distributive justice. The framework requires that utilities be measurable or at least comparable to enable aggregation and maximization.

Typical failure modes or objections to utilitarianism include the demandingness objection (maximization is overly demanding), the separateness of persons objection (aggregation ignores individual rights and separateness), the swine objection (hedonism reduces human goods to base pleasures), the problem of interpersonal utility comparability, and the problem of moral motivation.

The framework distinguishes between act utilitarianism, which applies the utility maximization principle directly to individual acts, and rule utilitarianism, which applies it to rules whose acceptance maximizes utility, thereby justifying adherence to rules even when breaking them in particular cases might maximize utility.

The framework's non-negotiables include the primacy of welfare maximization, impartial aggregation of utilities, and the consequentialist evaluation of actions or policies. It rejects moral codes based on tradition, authority, or intrinsic duties independent of consequences.

In contemporary debates, utilitarianism grapples with the problem of interpersonal comparability of utility. Some versions assume full comparability (cardinal full comparability), allowing direct comparisons of utility increments across individuals, while others adopt weaker comparability assumptions (cardinal non-comparability), which restrict meaningful comparisons to within individuals only.

The framework also acknowledges the need for decision procedures or social welfare functions that aggregate individual utilities into a social ranking, with axiomatic foundations ensuring properties like Pareto efficiency, independence, and continuity.

The framework is contested on grounds including the feasibility of utility measurement, the moral permissibility of sacrificing individual rights for aggregate welfare, and the plausibility of impartial aggregation in real-world social contexts.

3) Principle-by-Principle Deep Critical Evaluation

The evaluation of the welfarist curtailment use-case through the lens of utilitarianism requires a detailed examination of the core principles of the framework and their instantiation or omission in the use-case.

First, the use-case's explicit adoption of a welfarist perspective aligns fundamentally with utilitarianism's commitment to welfare as the morally primary value. The use-case treats prosumer utilities—proxied by power generation or export—as the relevant welfare metric, and seeks to maximize a social welfare function derived from these utilities. This is consistent with utilitarianism's aggregation principle, which requires that individual utilities be combined into a social ranking to guide moral or policy decisions.

However, the use-case departs from classical utilitarianism's common assumption of full interpersonal comparability of utilities. Instead, it adopts cardinal non-comparability (CNC), recognizing that utilities are cardinally meaningful within agents but not comparable across agents. This is a significant divergence from classical utilitarianism, which typically assumes that utilities can be summed across individuals on a common scale. The use-case justifies this stance on empirical and practical grounds: in heterogeneous residential systems, full comparability is often untenable, and assuming it can lead to undesirable outcomes.

From the standpoint of utilitarianism, this divergence is both a strength and a limitation. It is a strength because it reflects a realistic acknowledgment of the epistemic and normative difficulties in interpersonal utility comparisons, thus avoiding the pitfalls of arbitrary or unjustified aggregation. It is a limitation because classical utilitarianism's commitment to aggregation presupposes some degree of interpersonal comparability; without it, the standard utilitarian calculus is undermined. The use-case addresses this by introducing normative reference points (fallback and utopia envelopes) to anchor utility comparisons, enabling the definition of the Kalai-Smorodinsky (KS) social welfare function, which maximizes the minimum relative gain across agents.

This solution is innovative and aligns with utilitarianism's commitment to impartiality and inclusivity by ensuring that no agent's relative gain is sacrificed for others. The KS solution respects the aggregation of welfare in a way consistent with CNC, preserving the utilitarian ideal of maximizing social welfare while accommodating the epistemic constraints on interpersonal comparisons.

The use-case also adheres to utilitarianism's principle of consequentialism. The curtailment decision is modeled as an optimization problem over feasible operating points, with the social welfare function serving as the objective. This formalization ensures that decisions are evaluated solely based on their consequences for prosumer utilities, consistent with utilitarianism's rejection of intrinsic duties or rights independent of consequences.

Moreover, the use-case's axiomatic foundation, invoking Weak Pareto efficiency, Independence of Irrelevant Alternatives, Continuity, and Anonymity, reflects the utilitarian commitment to rational, consistent, and impartial social choice. The adherence to Weak Pareto ensures that decisions improving all agents' utilities are preferred, aligning with utilitarianism's maximization principle. Anonymity enforces impartiality by treating agents symmetrically.

However, the use-case's relaxation of IIA to allow reference points introduces a normative element that is not purely consequentialist. The selection of fallback and utopia envelopes involves explicit design choices reflecting normative judgments about baseline entitlements and ideal outcomes. While utilitarianism allows for such normative judgments insofar as they affect the welfare calculus, the use-case's reliance on these references to circumvent impossibility results indicates a departure from pure act utilitarianism's direct consequentialism.

This reliance on normative reference points can be interpreted as a form of rule utilitarianism or constrained utilitarianism, where rules or baselines guide the aggregation of utilities. This is consistent with the framework's recognition of rule utilitarianism as a viable variant addressing some classical utilitarianism's limitations.

The use-case's unification of existing curtailment schemes as instances of the KS solution further demonstrates its alignment with utilitarianism's principle of maximizing social welfare under constraints. By showing that various fairness criteria correspond to different normative reference points within the KS framework, the use-case respects the utilitarian ideal of rationally justifying social choices through welfare maximization while accommodating diverse fairness intuitions.

Nevertheless, the use-case's proxying of utility by power generation or export metrics raises concerns about the adequacy of welfare measurement, a perennial challenge for utilitarianism. The framework requires that utilities be measurable or at least comparable to enable aggregation and maximization. The use-case acknowledges that true agent utilities are not directly measurable and must be proxied, but it does not elaborate on the normative or empirical justification for these proxies as accurate reflections of welfare.

From a utilitarian perspective, this under-specification is significant. If the proxies poorly capture actual welfare, the social welfare function's maximization may not correspond to genuine utility maximization, undermining the moral legitimacy of the curtailment decisions. The use-case's transparency and auditability partly mitigate this concern, as explicit design choices can be scrutinized and adjusted, but the lack of a robust welfare measurement framework remains a limitation.

Furthermore, the use-case's focus on fairness as equal relative gains from fallback to utopia aligns with utilitarianism's impartiality and inclusivity but may conflict with the framework's classical maximization requirement. The KS solution maximizes the minimum relative gain, which is a form of maximin fairness rather than total utility maximization. While this approach ensures no agent is left behind, it may sacrifice aggregate welfare, raising the demandingness and efficiency trade-offs familiar in utilitarian debates.

The use-case does not explicitly address these trade-offs or justify the choice of the KS solution over utilitarian total welfare maximization. This omission leaves open the question of whether the approach fully satisfies utilitarianism's maximization principle or represents a compromise with egalitarian concerns, akin to a mixed utilitarian-egalitarian approach.

Finally, the use-case's rejection of full interpersonal comparability and adoption of CNC reflects a cautious epistemic stance but may limit the scope of utilitarian aggregation. The framework recognizes that interpersonal comparability assumptions are contested and that weaker comparability assumptions complicate social welfare aggregation. The use-case's solution via reference points and the KS function is a pragmatic response but may not fully resolve the theoretical tensions inherent in utilitarian aggregation without interpersonal comparability.

In summary, the use-case aligns with utilitarianism's core commitments to welfare as the moral metric, consequentialist evaluation, impartiality, and inclusivity. It innovatively addresses the interpersonal comparability problem through normative reference points and the KS social welfare function, providing a principled, axiomatic foundation for fair curtailment decisions. However, it diverges from classical utilitarianism's full comparability and maximization assumptions, relies on normative design choices that introduce rule-like constraints, and under-specifies the welfare measurement problem. These divergences are not necessarily fatal but represent significant theoretical and practical challenges from a utilitarian standpoint.

4) Stress Tests / Edge Cases

To further probe the robustness of the use-case under utilitarian scrutiny, consider two hypothetical scenarios that stress-test its principles.

First, imagine a scenario where a prosumer invests heavily in local generation upgrades, thereby increasing their utopia envelope substantially relative to others. The KS solution's Individual Monotonicity axiom requires that this prosumer's utility should not decrease as a result of the expanded feasible set. However, in a highly constrained grid, increased capacity by one prosumer may necessitate greater curtailment of others to maintain feasibility, potentially reducing their utilities. While the KS solution ensures monotonicity for the investing prosumer, it may lead to significant relative losses for others. From a utilitarian perspective, this raises questions about the balance between individual incentives and aggregate welfare. The use-case's reliance on relative gains may inadequately capture the overall welfare impact, especially if the losses to others outweigh the gains to the investor, challenging the framework's maximization principle.

Second, consider a scenario where the fallback envelope is set too optimistically, for example, assuming a baseline level of generation or export that is infeasible for some prosumers due to local constraints or outages. This mis-specification would distort the relative gains calculation in the KS function, potentially privileging some prosumers unfairly and violating Weak Pareto efficiency by preferring allocations that do not improve all agents' welfare relative to their true baseline. Such a scenario reveals the sensitivity of the use-case's normative reference points and the risk that their mis-specification undermines fairness and efficiency. From a utilitarian standpoint, this highlights the importance of accurate welfare baselines to ensure that social welfare maximization corresponds to genuine improvements in well-being.

These edge cases reveal that while the use-case's axiomatic approach provides a rigorous foundation, its practical application depends critically on normative design choices and accurate welfare measurement. The potential for trade-offs between individual incentives and aggregate welfare, and the sensitivity to baseline specifications, represent principled challenges that require careful management to maintain alignment with utilitarian principles.

5) Recommendations

In light of the above evaluation, several recommendations emerge to enhance the use-case's alignment with utilitarianism.

First, the use-case should explicitly address the welfare measurement problem by developing or integrating robust methods to validate that the chosen utility proxies (power generation, export) accurately reflect prosumer well-being. This could involve empirical studies linking these proxies to prosumer satisfaction, economic benefits, or other welfare indicators. Clarifying and justifying these proxies would strengthen the moral legitimacy of the social welfare function maximization.

Second, the normative design choices regarding fallback and utopia envelopes should be made transparent, participatory, and subject to periodic review. Given their critical role in anchoring interpersonal utility comparisons and shaping fairness outcomes, involving stakeholders in their determination would enhance the ethical robustness and social acceptability of curtailment policies.

Third, the use-case should explicitly engage with the trade-offs between egalitarian fairness (maximizing minimum relative gains) and utilitarian maximization of total welfare. This includes acknowledging that the KS solution prioritizes equity potentially at the expense of aggregate welfare and providing normative justification for this choice within the utilitarian framework, possibly by appealing to rule utilitarianism or mixed utilitarian-egalitarian theories.

Fourth, the use-case could consider integrating dynamic or longitudinal welfare considerations, recognizing that curtailment decisions have cumulative effects over time. Incorporating temporal aggregation of utilities would align with utilitarianism's commitment to maximizing overall welfare and address concerns about fairness and incentives in the long run.

Finally, the use-case should rigorously test and validate the sensitivity of outcomes to the specification of reference points and utility proxies, ensuring that fairness and efficiency are robust to reasonable variations. This would mitigate risks of mis-specification undermining utilitarian commitments and provide confidence in the ethical soundness of the curtailment decisions.

In conclusion, the use-case represents a sophisticated and principled application of utilitarian ethics to a complex socio-technical problem. Its innovations in addressing interpersonal comparability and fairness are commendable. However, to fully satisfy utilitarianism's core principles, it must strengthen its welfare measurement foundations, clarify normative choices, and explicitly engage with the inherent trade-offs between equity and efficiency.

Rawlsian Justice Framework evaluation

ALIGNMENT	CONCERN	CONFIDENCE	BENCHMARK
65	45	80	**Mixed** The use-case aligns well with Rawlsian principles of fairness, transparency, and concern for the least advantaged through the KS rule, but diverges on political equality, legitimacy, and institutional participation. Its technical focus limits full realization of Rawlsian justice, though many gaps could be addressed with structural enhancements.

1) The Use-Case, Reconstructed

The use-case under evaluation is a proposed welfarist framework for fair active power curtailment in distribution grids with distributed photovoltaics (PV). The decision problem addressed is how grid operators should allocate limited active power generation capacity among multiple prosumers—households or entities that both consume and generate electricity—when grid constraints prevent all from injecting their full potential power simultaneously. This problem arises due to physical limitations such as voltage, line current, and transformer capacity, which necessitate curtailment of some generation to maintain grid stability and safety.

The mechanism proposed is to model the curtailment decision as a social choice problem over the set of feasible operating points, where each operating point corresponds to a vector of allowed generation envelopes for all prosumers. The core innovation is to ground the curtailment objective in a social welfare function (SWF) derived from foundational axioms reflecting fairness and grid access rights, rather than relying on ad hoc or heuristic fairness metrics. The framework adopts a cardinal non-comparability stance on utilities, meaning that while utilities are cardinally meaningful within each agent's preferences, they are not assumed to be comparable across agents. This stance requires fewer assumptions about prosumers' private preferences and avoids problematic interpersonal utility comparisons.

The key social welfare function identified as uniquely satisfying the axioms under this stance is the Kalai-Smorodinsky (KS) rule. The KS rule maximizes the minimum relative gain of each prosumer's utility from a defined fallback (worst-case) envelope to a utopia (ideal) envelope, ensuring a form of proportional fairness that respects individual entitlements and potential. The framework further unifies existing curtailment schemes in the literature by showing that they correspond to KS solutions with different normative choices of fallback and utopia points.

The target outcomes are multiple: to provide a principled, axiomatic foundation for fairness in curtailment decisions; to improve transparency and auditability of curtailment policies; to ensure non-discriminatory and equitable access to the grid; to maintain system-wide efficiency by avoiding Pareto-dominated allocations; and to incentivize investment in renewable generation by ensuring that improvements in individual capabilities do not lead to worse outcomes.

Implicit value claims include a commitment to fairness understood as equitable sharing of a public resource (the grid), respect for individual rights to self-consumption and export, and social equity that balances efficiency with distributive justice. The framework privileges a procedural and axiomatic justification of fairness over utilitarian aggregation or arbitrary indices, emphasizing the importance of respecting the diversity of prosumers' utilities without forcing interpersonal comparisons.

The decision context involves multiple stakeholders: prosumers who seek to maximize their generation and economic returns; grid operators who must maintain technical constraints and regulatory compliance; regulators who mandate non-discriminatory access and fairness; and society at large, which values sustainability, investment promotion, and equitable resource distribution. Success is measured by the ability to select curtailment profiles that are feasible, fair according to the axiomatic criteria, efficient in terms of grid utilization, and transparent in justification.

The use-case assumes the availability of utility proxy metrics (e.g., generated power or export) for each prosumer, though it acknowledges that true utilities are not directly measurable. It also assumes convex and monotonic grid constraints, allowing for tractable optimization. The framework is designed for static, snapshot decisions but suggests potential extensions to dynamic, time-series settings.

2) The Framework, Reconstructed

The ethical framework for evaluation is Rawlsian Justice, as elaborated in the comprehensive overview provided. Rawlsian Justice is a liberal egalitarian political philosophy centered on the principles of justice as fairness. Its core commitments include treating all citizens as free and equal moral persons, ensuring that social institutions are arranged to benefit all, especially the least advantaged, and securing equal basic liberties for all.

Rawlsian Justice emphasizes the importance of legitimacy and stability in political arrangements, achieved through principles that all reasonable citizens can endorse despite pluralism of worldviews. The framework distinguishes between ideal and non-ideal theory, focusing primarily on ideal theory that assumes compliance and favorable conditions to establish principles of justice. It employs the method of reflective equilibrium to justify principles through coherence among considered judgments and theoretical commitments.

The framework's normative core comprises two principles of justice: (1) equal basic liberties for all, which have lexical priority, and (2) social and economic inequalities arranged so that they are attached to positions open to all under fair equality of opportunity and benefit the least advantaged (the difference principle). Citizens are conceived as free, equal, reasonable, and rational agents with two moral powers: a capacity for a sense of justice and a capacity for a conception of the good.

Rawlsian Justice treats social cooperation as fundamental and demands that institutions be publicly justifiable, transparent, and stable through overlapping consensus. It rejects utilitarianism's aggregation of welfare at the expense of individual rights and fairness. The framework also requires that political power be exercised according to principles that all citizens can reasonably endorse, respecting reciprocity and public reason.

Typical failure modes of the framework include arrangements that violate equal basic liberties, fail to secure fair equality of opportunity, or produce inequalities that do not benefit the least advantaged. It also critiques social orders that impose comprehensive doctrines or fail to accommodate reasonable pluralism.

Rawlsian Justice treats justice as the primary moral value in structuring society's basic institutions, with a strong emphasis on fairness, reciprocity, and respect for persons as free and equal. It demands that distributive principles be justified from an impartial standpoint (the original position) behind a veil of ignorance, ensuring that arbitrary contingencies do not bias the principles.

Objections to Rawlsian Justice often concern its idealizing assumptions, the feasibility of overlapping consensus, and the scope of its egalitarian commitments. However, these do not undermine its normative force as a framework for evaluating justice in social institutions.

3) Principle-by-Principle Deep Critical Evaluation

Evaluating the welfarist curtailment framework through the lens of Rawlsian Justice requires careful mapping of the framework's features onto Rawls's core principles and conceptions of justice, fairness, and political legitimacy.

Equal Basic Liberties and Rights

Rawls's first principle of justice demands that all citizens have equal and indefeasible basic liberties, including political liberties and rights to participate in social cooperation. In the context of the use-case, this translates into ensuring that all prosumers have equal rights to access the grid and to participate in energy generation and distribution without arbitrary discrimination.

The use-case explicitly addresses fair grid access as a key criterion, emphasizing non-discriminatory treatment and equitable sharing of the public resource. The adoption of a social welfare function that respects individual entitlements (fallback envelopes) and potential (utopia envelopes) aligns with Rawls's insistence on equal basic liberties insofar as these define the minimal rights and opportunities each prosumer should have.

However, Rawls's framework requires not only formal equality but also the fair value of political liberties, meaning that inequalities in influence or access due to wealth or power must be corrected. The use-case does not explicitly address political equality or the distribution of power among prosumers in decision-making processes. It focuses on the distribution of physical grid access rather than on the political or institutional structures governing that access.

This omission is significant because Rawlsian Justice would require that the institutions managing curtailment ensure that all prosumers have fair political opportunities to influence rules and policies, preventing capture by wealthier or more powerful agents. The use-case's focus on technical fairness in curtailment, while necessary, is insufficient to satisfy the full scope of Rawls's first principle.

Fair Equality of Opportunity

Rawls's second principle prioritizes fair equality of opportunity, requiring that offices and positions be open to all under conditions that neutralize the effects of morally arbitrary factors such as social class or natural endowments.

The use-case's cardinal non-comparability stance on utilities reflects a sensitivity to the diversity of prosumers' circumstances and preferences, avoiding unjustified interpersonal comparisons. By defining fallback and utopia envelopes tailored to each prosumer's potential and rights (e.g., self-consumption as a fallback), the framework respects differences in starting points and capabilities.

This approach resonates with Rawls's requirement that social institutions compensate for arbitrary inequalities to ensure fair opportunities. For example, the proportional redistribution scheme equalizes export shares relative to each prosumer's potential, which can be seen as a form of compensating for differences in capacity.

Nevertheless, Rawlsian Justice demands more than proportional fairness; it requires that inequalities be arranged to benefit the least advantaged. The use-case's reliance on the KS rule maximizes the minimum relative gain, which is a strong egalitarian criterion aligned with benefiting the worst-off in the feasible set. This is a significant point of convergence.

However, the use-case does not explicitly identify or prioritize the least advantaged prosumers in a social or economic sense beyond the mathematical minimization of relative gains. Rawls's difference principle is grounded in a moral conception of disadvantage that includes social status, economic power, and life prospects, not merely utility proxies.

The use-case's abstraction to utility proxies such as power generation or export may fail to capture deeper social disadvantages or barriers that Rawlsian Justice would require addressing. For example, a prosumer with limited economic means or social capital might still face systemic disadvantages not reflected in the utility proxy.

Thus, while the KS rule's mathematical structure aligns with Rawls's difference principle in form, the use-case lacks a substantive account of who counts as least advantaged and how curtailment decisions affect their broader social position.

Reciprocity and Reasonable Pluralism

Rawlsian Justice emphasizes reciprocity and the reasonable pluralism of citizens' comprehensive doctrines. Political power and social cooperation must be justified to all citizens as free and equal, who hold diverse worldviews.

The use-case's welfarist framework is neutral with respect to prosumers' comprehensive doctrines or values, focusing instead on utility proxies and axiomatic fairness. This neutrality can be interpreted as respecting reasonable pluralism by not imposing a particular conception of the good beyond the minimal assumption that more utility is better.

Moreover, the use-case's axiomatic approach provides a common framework for justifying curtailment decisions to all prosumers, enhancing legitimacy through transparency and auditability. This procedural fairness supports the Rawlsian ideal of public reason and legitimacy.

However, Rawlsian Justice requires that the principles of justice be publicly justifiable and stable through overlapping consensus. The use-case does not address how prosumers with divergent values or priorities might accept the KS-based curtailment scheme as legitimate or fair. It assumes acceptance of the utility proxies and axioms without exploring the social or political processes that would secure overlapping consensus.

This gap is critical because Rawlsian legitimacy depends on citizens' reasonable endorsement of the principles governing coercive power—in this case, the power to curtail generation. The use-case's technical and axiomatic rigor does not guarantee such endorsement, especially if prosumers perceive the utility proxies or fallback/utopia points as arbitrary or misaligned with their values.

Publicity and Transparency

Rawlsian Justice requires that principles of justice and their justifications be publicly accessible and transparent, enabling citizens to understand and endorse them.

The use-case explicitly aims to improve transparency and auditability by grounding curtailment decisions in axiomatic social choice theory. By unifying existing schemes under the KS framework and making the choice of fallback and utopia points explicit design decisions, it enhances the publicity of the decision-making process.

This feature aligns well with Rawls's publicity condition and duty of civility in public reason. The use-case's rejection of ad hoc metrics in favor of principled, explainable objectives supports the Rawlsian ideal of a well-ordered society where institutions are not shrouded in ideology or manipulation.

Stability and Social Cooperation

Rawlsian Justice demands that principles of justice be stable over time, supported by citizens' sense of justice and mutual trust.

The use-case does not explicitly address stability in the Rawlsian sense, as it focuses on static curtailment decisions rather than the broader social and political institutions that sustain cooperation. However, by maximizing the minimum relative gain and respecting fallback entitlements, the KS rule fosters a form of reciprocity and mutual benefit that could contribute to social stability among prosumers.

Yet, without mechanisms for political participation, dispute resolution, or institutional enforcement that Rawlsian Justice requires, the framework's contribution to stability is limited. The use-case's technical optimization does not guarantee that prosumers will accept curtailment decisions as legitimate or just, especially over repeated interactions.

The Original Position and Impartiality

Rawlsian Justice's hallmark is the original position, a hypothetical impartial standpoint behind a veil of ignorance that ensures fairness by removing knowledge of arbitrary contingencies.

The use-case's adoption of the KS rule with fallback and utopia points can be interpreted as a partial analogue to the original position's impartiality. By normalizing utilities relative to worst-case and best-case benchmarks, it abstracts from individual circumstances to a common scale.

However, the use-case does not explicitly model or justify these reference points through a veil of ignorance or impartial deliberation. The choice of fallback and utopia points remains a normative design decision without a formal mechanism for impartial agreement among prosumers.

This under-specification weakens the framework's claim to embody Rawlsian impartiality. Without a procedure akin to the original position, the legitimacy and fairness of the chosen reference points may be contested.

Political Constructivism and Justification

Rawls's political constructivism holds that principles of justice are justified through public reasoning from a shared conception of citizens and society.

The use-case's axiomatic derivation of the KS social welfare function aligns with the spirit of political constructivism by justifying curtailment decisions through shared principles rather than contingent preferences.

Yet, the use-case does not engage with the broader political and moral reasoning that Rawlsian constructivism entails. It abstracts from the social meanings of energy generation and distribution, and from the political institutions that govern grid access.

Consequently, while the use-case's axiomatic rigor is commendable, it falls short of the full justificatory demands of Rawlsian political constructivism.

Summary of Alignment and Divergence

In summary, the use-case aligns with Rawlsian Justice in its commitment to fairness as equitable sharing of a public resource, its procedural transparency, and its use of principles that maximize the minimum relative gain, echoing the difference principle's concern for the least advantaged. Its cardinal non-comparability stance respects the diversity of prosumers' utilities without unjustified interpersonal comparisons.

However, it diverges or remains under-justified in several critical respects: it neglects the political dimension of equal basic liberties and fair equality of opportunity; it lacks an explicit mechanism for securing legitimacy and overlapping consensus; it does not incorporate the full institutional and political context Rawlsian Justice requires; and it under-specifies the normative basis for choosing fallback and utopia points, weakening impartiality.

These divergences are partly intrinsic, given the use-case's technical and operational focus, but some could be addressed through deeper institutional design and participatory processes.

4) Stress Tests / Edge Cases

To probe the robustness of the use-case under Rawlsian Justice, consider two hypothetical scenarios:

Scenario 1: A Prosumer with Limited Economic Means but High Potential Generation

Suppose a prosumer lives in a low-income household with a high-capacity PV installation but limited ability to invest in grid upgrades or political advocacy. The utility proxy (e.g., export power) is high potential, but the prosumer faces systemic disadvantages in influencing grid policies or accessing subsidies.

Under the use-case, the KS rule would allocate curtailment based on relative gains from fallback to utopia, potentially granting this prosumer a fair share of generation capacity. However, Rawlsian Justice would require attention to the prosumer's broader social disadvantage, ensuring fair equality of opportunity not only in generation but also in political participation and economic resources.

The use-case's abstraction to utility proxies misses this dimension, risking perpetuation of structural inequalities. Without institutional mechanisms to address these disadvantages, the prosumer's equal access to grid capacity may be insufficient to achieve justice.

This scenario reveals a principled flaw: the use-case's limited scope cannot fully realize Rawlsian demands for fair equality of opportunity and political equality.

Scenario 2: Disagreement Among Prosumers on the Definition of Utility and Fairness

Imagine a community where prosumers have divergent values: some prioritize maximizing self-consumption, others prioritize exporting for economic gain, and others value environmental sustainability differently. They disagree on what utility proxy best represents their interests and on what fallback and utopia points should be.

The use-case assumes a common utility proxy and normative reference points chosen by the designer. Rawlsian Justice requires that principles of justice be publicly justifiable and stable through overlapping consensus among reasonable citizens with diverse worldviews.

In this scenario, the use-case's lack of a participatory or deliberative process to select utility proxies and reference points undermines its legitimacy and stability. Prosumers may reject the imposed framework as failing to respect their comprehensive doctrines or reasonable pluralism.

This scenario highlights the use-case's under-specification of political legitimacy and public reason, critical for Rawlsian Justice.

5) Recommendations

In light of the evaluation, several recommendations emerge to improve alignment with Rawlsian Justice:

First, the use-case should explicitly incorporate political and institutional dimensions of fairness, ensuring that all prosumers have fair opportunities to participate in decision-making about curtailment policies. This would address Rawls's requirement of fair equality of opportunity and the fair value of political liberties.

Second, the framework should be extended to include mechanisms for public justification and deliberation, enabling prosumers to endorse the choice of utility proxies, fallback, and utopia points. This would enhance legitimacy through overlapping consensus and respect reasonable pluralism.

Third, the normative basis for selecting fallback and utopia envelopes should be grounded in a procedure analogous to Rawls's original position, ensuring impartiality and fairness in the design of curtailment objectives.

Fourth, the framework should expand the conception of disadvantage beyond utility proxies to include social and economic factors affecting prosumers' life prospects and political power, aligning with the difference principle's substantive egalitarianism.

Fifth, the framework should consider dynamic and institutional stability over time, incorporating mechanisms to sustain trust and cooperation among prosumers beyond static optimization.

These revisions would require structural changes to the use-case, moving beyond technical optimization to integrated socio-technical governance. While some divergences are intrinsic to the use-case's operational focus, addressing these recommendations would significantly enhance its normative robustness under Rawlsian Justice.

Rights-Based Ethics Framework evaluation

ALIGNMENT	CONCERN	CONFIDENCE	BENCHMARK
55	65	70	**Mixed** The use case demonstrates some alignment with Rights-Based Ethics through its emphasis on fairness and equitable access, but it diverges in critical areas related to the identification of rights and the handling of competing claims.

Framework Evaluation of A Welfarist Perspective on Fair Generation Curtailment via Rights-Based Ethics

1) The Use-Case, Reconstructed

The use case presents a welfarist approach to fair active power curtailment in distribution grids with distributed photovoltaics (PV). The decision problem at hand involves determining which generators to curtail in order to respect grid operational constraints while balancing multiple criteria such as sustainability, fair grid access, social equity, and investment promotion. The authors argue that existing curtailment rules often lack consistent axiomatic foundations, leading to undesirable outcomes like inefficiency and disincentivization of future investments.

The proposed mechanism involves modeling the curtailment decision as a social choice problem over feasible operating points and deriving curtailment objectives from foundational axioms that express principled stances on fairness and grid access rights. The authors utilize the Kalai-Smorodinsky (KS) social welfare function to create a framework that allows for fair social ranking of curtailment decisions.

The target outcomes of this approach are to ensure equitable access to the grid for all prosumers while minimizing total curtailment of renewable sources. The authors assert that their method provides a transparent, auditable foundation for justifying fairness in local energy systems.

Implicit value claims in the use case include the prioritization of fairness and equity in energy distribution, the necessity of sustainability in energy practices, and the importance of promoting investments in renewable energy. The authors assume that fairness can be quantitatively assessed and operationalized through the KS social welfare function, which is intended to balance the needs of various stakeholders, including prosumers, grid operators, and society at large.

The use case also implies that the current regulatory and operational frameworks are insufficient for addressing the complexities introduced by distributed energy resources. By advocating for a principled approach, the authors suggest that existing curtailment schemes can be unified under a common framework, thus enhancing both transparency and efficiency.

Success in this use case is defined by the ability to achieve a fair distribution of curtailment burdens among prosumers while maintaining grid stability and encouraging future investments in renewable energy. The authors assert that their approach will lead to a more equitable energy landscape, where all stakeholders have a fair opportunity to contribute to and benefit from the energy system.

2) The Framework, Reconstructed

Rights-Based Ethics, or Rights Theory, emphasizes the centrality of rights as moral entitlements that constrain how individuals and institutions may act toward one another. This framework posits that certain individuals possess entitlements—freedoms, claims, protections, powers, and immunities—that must be respected by others. The core principles of Rights-Based Ethics include the recognition of individual rights, the obligation of duty-bearers to respect those rights, and the importance of institutional mechanisms to enforce these rights.

The framework operates under several non-negotiables. First, it distinguishes between legal rights and moral rights, asserting that moral rights exist independently of legal recognition. Second, it emphasizes the importance of identifying duty-bearers—those who have obligations to uphold the rights of others—and the nature of those duties, which may include respect, protection, facilitation, and provision.

Rights-Based Ethics also recognizes the potential for conflicts between rights and the necessity of balancing competing rights claims. This balancing act must be approached with caution, as it can lead to the undermining of the very rights that the framework seeks to protect. The framework is often critiqued for its potential to generate rigid moral absolutes, which may not account for the complexities of real-world situations where rights may conflict.

The moral primacy of Rights-Based Ethics lies in its commitment to individual autonomy and dignity. It seeks to protect individuals from abuse and domination, ensuring that their rights are not overridden by utilitarian calculations or majority preferences. Critics of the framework argue that it can lead to an adversarial discourse that emphasizes individual claims over communal responsibilities, potentially obscuring the moral duties owed to others.

In summary, Rights-Based Ethics provides a robust framework for evaluating moral claims based on the recognition and protection of individual rights. It emphasizes the need for institutional mechanisms to uphold these rights and navigates the complexities of rights conflicts with a focus on individual dignity and autonomy.

3) Principle-by-Principle Deep Critical Evaluation

The use case aligns with several principles of Rights-Based Ethics, particularly in its emphasis on fairness and equitable access to grid resources. The authors assert that their welfarist approach is grounded in principles of fairness, which resonates with the rights-based commitment to equitable treatment of individuals. The focus on minimizing total curtailment of renewable sources also aligns with the rights-based emphasis on protecting individual entitlements to generate and consume energy.

However, the use case diverges from the framework in several critical areas. First, while the authors advocate for a fair distribution of curtailment burdens, they do not explicitly identify the rights of prosumers in relation to grid access. The framework requires a clear articulation of who the duty-bearers are and what specific rights are being upheld. The absence of this explicit identification raises questions about the moral justification of the proposed curtailment schemes.

Second, the use case relies heavily on the Kalai-Smorodinsky social welfare function as a mechanism for achieving fairness. While this approach may provide a quantitative measure of fairness, it risks reducing complex moral claims to mere mathematical calculations. Rights-Based Ethics emphasizes the importance of moral reasoning and the recognition of individual rights, which may not be adequately captured by a purely welfarist framework. The reliance on a social welfare function may obscure the individual rights of prosumers and their entitlements, leading to potential violations of those rights.

Furthermore, the use case does not adequately address the potential conflicts that may arise between the rights of different prosumers. Rights-Based Ethics requires a careful consideration of how competing rights claims will be balanced, particularly in a context where some prosumers may have more substantial claims to grid access than others. The authors' failure to engage with this complexity raises concerns about the robustness of their proposed solutions.

The use case also implies a certain level of interpersonal comparability among prosumers' utilities, which may not align with the rights-based commitment to individual autonomy. Rights-Based Ethics emphasizes that rights are not merely about utility maximization but are grounded in the inherent dignity and worth of each individual. The assumption that prosumers' utilities can be compared and aggregated may undermine the moral significance of their individual rights.

In summary, while the use case demonstrates some alignment with Rights-Based Ethics through its emphasis on fairness and equitable access, it diverges in critical areas related to the identification of rights, the potential reduction of moral claims to quantitative measures, and the handling of competing rights claims. These divergences highlight the need for a more nuanced engagement with the principles of Rights-Based Ethics to ensure that the proposed solutions genuinely uphold the rights and dignity of all stakeholders involved.

4) Stress Tests / Edge Cases

To further evaluate the robustness of the use case under Rights-Based Ethics, it is essential to consider potential stress tests or edge cases that may reveal underlying tensions or flaws in the proposed approach.

One scenario involves a situation where a significant imbalance exists among prosumers in terms of their energy generation capacity and consumption needs. For instance, consider a case where one prosumer has a substantial PV installation capable of generating excess energy, while another prosumer has minimal capacity and relies heavily on grid access for their energy needs. In this scenario, the application of the Kalai-Smorodinsky social welfare function may lead to a curtailment decision that disproportionately affects the prosumer with limited generation capacity, thereby undermining their rights to access energy. This situation raises critical questions about the fairness of the curtailment decision and whether it adequately respects the rights of all prosumers involved.

Another edge case could involve a situation where external factors, such as regulatory changes or market fluctuations, significantly alter the operational constraints of the grid. For example, if new regulations mandate a higher percentage of renewable energy integration, the curtailment decisions may need to shift rapidly to comply with these requirements. In such a case, the reliance on a fixed social welfare function may prove inadequate, as it may not account for the dynamic nature of energy markets and the evolving rights of prosumers. This scenario highlights the need for a more flexible approach that can adapt to changing circumstances while still upholding the rights and entitlements of all stakeholders.

Both scenarios illustrate the potential for conflicts between the rights of different prosumers and the need for a more nuanced understanding of fairness that goes beyond mere utility calculations. They underscore the importance of explicitly identifying the rights of prosumers and the duty-bearers responsible for upholding those rights. The use case's current framework may struggle to navigate these complexities, revealing intrinsic limitations in its approach to fairness and rights.

5) Recommendations

In light of the evaluation, several recommendations can be made to enhance the alignment of the use case with Rights-Based Ethics.

First, it is crucial to explicitly identify the rights of prosumers in relation to grid access and energy generation. This identification should include a clear articulation of the entitlements that each prosumer holds and the corresponding duties of grid operators and other stakeholders to respect those rights. By grounding the approach in a clear understanding of rights, the authors can better navigate potential conflicts and ensure that all prosumers are treated equitably.

Second, the use case should incorporate a more qualitative assessment of fairness that goes beyond quantitative measures. While the Kalai-Smorodinsky social welfare function provides a useful framework for social ranking, it risks reducing complex moral claims to mere calculations. Engaging with the moral significance of individual rights and the inherent dignity of prosumers can enrich the analysis and lead to more ethically robust curtailment decisions.

Third, the framework should be flexible enough to adapt to changing circumstances and evolving rights claims. This adaptability could involve incorporating mechanisms for ongoing stakeholder engagement and feedback, allowing for a more responsive approach to curtailment decisions that respects the rights of all prosumers.

Finally, the authors should consider the implications of interpersonal comparability among prosumers' utilities. Acknowledging the limitations of comparing utilities can help reinforce the rights-based commitment to individual autonomy and dignity. By emphasizing the uniqueness of each prosumer's situation, the authors can better align their approach with the principles of Rights-Based Ethics.

In conclusion, while the use case presents a promising approach to fair generation curtailment, it requires significant revisions to fully align with Rights-Based Ethics. By explicitly identifying rights, incorporating qualitative assessments, ensuring adaptability, and acknowledging the limitations of utility comparisons, the authors can strengthen their framework and promote a more equitable energy landscape.

Contractualism Framework evaluation

ALIGNMENT	CONCERN	CONFIDENCE	BENCHMARK
70	60	75	**Moderate Alignment** The use case aligns with several core principles of contractualism, particularly in its focus on fairness and reasonable rejection. However, it struggles with interpersonal comparability and conflict resolution, leading to meaningful concerns about equity and justifiability.

Framework Evaluation of A Welfarist Perspective on Fair Generation Curtailment via Contractualism

1) The Use-Case, Reconstructed

The use case under evaluation presents a welfarist approach to fair active power curtailment in distribution grids featuring distributed photovoltaics (PV). The decision problem arises from the operational constraints faced by grid operators when managing the influx of renewable energy sources, particularly solar power. As the integration of distributed PV systems increases, so does the necessity for curtailment to maintain grid stability and efficiency. The authors identify a lack of consistent axiomatic foundations in existing ad-hoc curtailment rules, which can lead to inefficiencies and inequities in how power is distributed among prosumers.

The proposed mechanism involves modeling the curtailment decision as a social choice problem over feasible operating points, deriving curtailment objectives from foundational axioms that express fairness and grid access rights. The authors argue for a shift from arbitrary fairness indices to a more principled, axiomatic approach grounded in social choice theory. By adopting a Kalai-Smorodinsky (KS) social welfare function, the authors aim to create a transparent and auditable framework for justifying fairness in local energy systems.

The target outcomes of this approach include minimizing total curtailment of renewable sources, ensuring fair grid access, promoting social equity, and encouraging investment in renewable generation. Success is defined not merely in terms of operational efficiency but also in achieving a fair distribution of curtailment burdens among prosumers. Implicit value claims include the importance of fairness, equity, and the recognition of prosumers as stakeholders with rights to their generated energy. The authors assume that prosumers have a legitimate claim to self-consumption and that equitable treatment in curtailment decisions is essential for fostering trust and participation in the energy transition.

The stakeholders involved in this decision-making process include grid operators, prosumers, regulatory bodies, and society at large, each with distinct interests and expectations. The optimization of curtailment decisions is framed within the context of social welfare, where the welfare of individual prosumers is aggregated to inform collective decision-making. The hidden assumptions in this use case revolve around the comparability of utilities among prosumers, the feasibility of implementing the proposed curtailment schemes, and the regulatory environment that governs grid access.

2) The Framework, Reconstructed

The ethical framework of contractualism, particularly as articulated by T. M. Scanlon, posits that moral principles are those that no one could reasonably reject as a basis for informed, unforced general agreement. Central to this framework is the idea of mutual recognition and respect for individuals as rational agents capable of making moral claims. Contractualism emphasizes the importance of justifiability to others, asserting that moral requirements derive their authority from the capacity for individuals to engage in rational discourse about the principles that govern their interactions.

Key principles of contractualism include:

1. **Reasonable Rejection**: A principle is only valid if it cannot be reasonably rejected by any individual affected by it. This criterion emphasizes the importance of individual perspectives and the necessity of justifying actions to all stakeholders involved.

2. **Mutual Recognition**: The framework underscores the value of recognizing each individual's capacity to assess reasons and justifications. This mutual recognition is foundational to establishing moral obligations.

3. **Impartiality**: Contractualism advocates for an impartial consideration of interests, which aligns with the notion of fairness in decision-making. Each individual's reasons for rejecting a principle must be taken into account without aggregation.

4. **Non-aggregation**: Unlike utilitarianism, contractualism does not permit the aggregation of individual interests. Each person's complaint must be assessed on its own merits, ensuring that no individual's rights are sacrificed for the greater good.

5. **Focus on Wrongness**: Contractualism prioritizes the concept of wrongness over rightness, asserting that an act is wrong if it cannot be justified to others. This focus helps delineate the moral landscape in which decisions are made.

The framework's non-negotiables include the commitment to reasonable rejection and mutual recognition, which serve as the foundation for moral reasoning. Typical failure modes involve scenarios where principles may be rejected due to perceived inequities or where the interests of marginalized groups are overlooked.

Critics of contractualism often point to its limitations in addressing issues of distributive justice and its reliance on the rationality of agents. Some argue that the framework may struggle to account for situations where individuals lack the capacity for rational deliberation or where power imbalances exist.

3) Principle-by-Principle Deep Critical Evaluation

Reasonable Rejection

The use case aligns with the principle of reasonable rejection by explicitly acknowledging the need for fairness in curtailment decisions. The authors advocate for a framework that allows prosumers to have their interests considered, thereby ensuring that no individual can reasonably reject the principles guiding curtailment. By modeling the decision as a social choice problem, the authors create a mechanism for evaluating curtailment schemes based on their justifiability to all affected parties.

However, the use case could be critiqued for not sufficiently addressing how the framework will handle situations where prosumers have conflicting interests. For instance, if one prosumer's need for self-consumption conflicts with another's desire for export, the framework must provide a clear mechanism for resolving such conflicts. The lack of explicit procedures for addressing these tensions may weaken the overall justification of the proposed curtailment schemes.

Mutual Recognition

The emphasis on mutual recognition is evident in the use case's focus on fairness and equity. The authors argue for a welfarist approach that respects the rights of prosumers to their generated energy. By framing the curtailment decision within a social welfare function, the authors aim to create a system that acknowledges the diverse interests of all stakeholders.

Nevertheless, the use case may fall short in fully operationalizing mutual recognition. While the authors advocate for fair treatment, the practical implementation of these principles may encounter challenges, particularly in heterogeneous residential systems where individual preferences and capacities may vary significantly. A more robust framework for mutual recognition would require a deeper exploration of how to balance these differences in a manner that is both fair and efficient.

Impartiality

The use case embodies the principle of impartiality by striving to create a fair distribution of curtailment burdens among prosumers. The authors propose a Kalai-Smorodinsky social welfare function that aims to maximize the minimum relative improvement for all agents involved. This approach reflects a commitment to treating all prosumers equitably, aligning with contractualism's emphasis on impartial consideration of interests.

However, the framework's reliance on interpersonal comparability raises concerns about the feasibility of achieving true impartiality. The assumption that utilities can be compared across different prosumers may not hold in practice, particularly when individual circumstances and preferences differ widely. This limitation could undermine the fairness of the proposed curtailment schemes, as it may inadvertently privilege certain prosumers over others based on arbitrary metrics.

Non-aggregation

The use case adheres to the principle of non-aggregation by rejecting the aggregation of individual utilities in favor of a focus on individual complaints. The authors emphasize that each prosumer's reasons for rejecting a principle must be considered independently, which aligns with contractualism's commitment to respecting individual rights.

However, the challenge remains in ensuring that this non-aggregation principle is upheld in practice. The authors must demonstrate how the proposed framework can effectively navigate situations where multiple prosumers may have conflicting interests. If the framework fails to adequately address these conflicts, it risks falling into the trap of aggregating interests in a manner that undermines the very principles it seeks to uphold.

Focus on Wrongness

The use case's focus on fairness and equity aligns with contractualism's emphasis on wrongness as a primary moral predicate. The authors argue that curtailment decisions must be justified to all affected parties, thereby framing the issue in terms of what is morally permissible. This focus on wrongness helps clarify the ethical stakes involved in curtailment decisions.

Nonetheless, the use case could benefit from a more explicit exploration of the moral implications of potential wrongness in curtailment decisions. For instance, the authors should consider the ethical ramifications of disproportionately burdening certain prosumers, even if the overall curtailment scheme is deemed fair. A more nuanced understanding of wrongness would enhance the framework's robustness and provide clearer guidance for decision-makers.

4) Stress Tests / Edge Cases

Scenario 1: Conflicting Interests Among Prosumers

Consider a situation where two prosumers, A and B, have conflicting interests regarding curtailment. Prosumer A relies heavily on self-consumption to meet their energy needs, while prosumer B seeks to maximize exports to the grid. If grid constraints necessitate curtailment, how should the framework resolve this conflict?

Under the current use case, the framework may struggle to provide a satisfactory resolution. While the principle of reasonable rejection emphasizes the need for justifiability to all parties, it does not offer a clear mechanism for addressing conflicting interests. This scenario reveals a potential flaw in the use case, as it may lead to arbitrary decisions that fail to respect the rights of both prosumers.

Scenario 2: Heterogeneous Preferences Among Prosumers

In a more complex scenario, imagine a distribution grid with multiple prosumers, each with unique preferences and capacities for energy generation. The proposed welfarist framework must account for these differences while maintaining fairness in curtailment decisions.

If the framework relies on a single social welfare function to aggregate utilities, it may inadvertently privilege certain prosumers whose preferences align more closely with the chosen metric. This could result in inequitable treatment of those with less common preferences, undermining the principle of mutual recognition. The framework must demonstrate how it can accommodate this heterogeneity without sacrificing fairness or efficiency.

5) Recommendations

To enhance the alignment of the use case with the principles of contractualism, several recommendations can be made.

First, the framework should incorporate explicit mechanisms for resolving conflicts among prosumers with competing interests. This could involve establishing a decision-making process that prioritizes mutual recognition and respects the rights of all stakeholders involved.

Second, the use case should address the challenges posed by interpersonal comparability. By acknowledging the limitations of comparing utilities across diverse prosumers, the framework can better uphold the principle of impartiality and ensure that all voices are heard in the decision-making process.

Third, the authors should provide a more nuanced exploration of the moral implications of curtailment decisions. By considering the ethical ramifications of disproportionately burdening certain prosumers, the framework can strengthen its focus on wrongness and provide clearer guidance for decision-makers.

Finally, the framework should be adaptable to accommodate the heterogeneous preferences of prosumers. This could involve developing a more flexible social welfare function that allows for the incorporation of diverse utility metrics, thereby ensuring that all prosumers are treated fairly and equitably.

In conclusion, while the proposed welfarist approach to fair generation curtailment demonstrates significant alignment with the principles of contractualism, addressing the identified weaknesses and enhancing the framework's robustness will be essential for achieving a truly fair and just energy system.

Principlism Framework evaluation

ALIGNMENT	CONCERN	CONFIDENCE	BENCHMARK
65	40	70	**Mixed** The use case aligns with several principles of Principlism, particularly in its emphasis on fairness and welfare. However, it lacks sufficient attention to potential harms and the mechanisms for ensuring prosumer autonomy, leading to meaningful concerns.

Framework Evaluation of A Welfarist Perspective on Fair Generation Curtailment via Principlism

1) The Use-Case, Reconstructed

The use case presented is a welfarist approach to fair active power curtailment in distribution grids with distributed photovoltaics (PV). The decision problem at hand involves how grid operators should manage the curtailment of power generated by prosumers—individuals or entities that both produce and consume electricity—when the grid cannot accommodate the total generation due to operational constraints. The proposed mechanism is a social choice model that derives curtailment objectives from foundational axioms expressing principled stances on fairness and grid access rights. The authors argue that existing curtailment rules lack consistent axiomatic foundations, leading to undesirable outcomes such as inefficiency and disincentivization of future investments.

The target outcomes of this approach are twofold: to establish a fair mechanism for curtailment that respects the rights of prosumers while ensuring grid stability and to provide an auditable, axiomatic foundation for justifying fairness in local energy systems. The authors suggest that the Kalai-Smorodinsky (KS) rule can serve as a social welfare function that meets these objectives.

Implicit value claims in the use case include the prioritization of fairness in the distribution of grid access rights, the importance of sustainability in energy production, and the need for transparency in decision-making processes. The stakeholders involved are diverse, including prosumers, grid operators, and regulatory bodies. The optimization focuses on balancing the needs of these stakeholders while adhering to operational constraints, with success measured by the equitable distribution of curtailment burdens and the preservation of grid integrity.

Hidden assumptions include the belief that prosumers have a right to a certain level of grid access, which may not be universally accepted. Additionally, the approach assumes that utilities can be adequately represented through a social welfare function, which may overlook the complexities of individual preferences and values. The use case also implies that the existing fairness metrics are insufficient, necessitating a more rigorous and principled approach to curtailment.

2) The Framework, Reconstructed

The ethical framework employed for this evaluation is Principlism, which is characterized by its reliance on four primary principles: respect for autonomy, nonmaleficence, beneficence, and justice. Principlism emerged as a practical response to the challenges of moral philosophy in bioethics, allowing for a common ground among diverse ethical theories. The framework emphasizes the importance of moral principles over specific moral traditions or complex philosophical theories, making it applicable to real-world ethical dilemmas.

Respect for autonomy requires that individuals have the freedom to make informed choices regarding their lives and circumstances. In the context of this use case, this principle would advocate for the rights of prosumers to have a say in how their energy production is managed and curtailed. Nonmaleficence, or the obligation to do no harm, would necessitate that curtailment practices do not adversely affect the welfare of prosumers or the broader community. Beneficence involves acting in the best interests of individuals and society, which in this case would mean ensuring that curtailment practices promote overall welfare and sustainability. Finally, justice pertains to the fair distribution of benefits and burdens, which is central to the proposed curtailment framework.

The framework's non-negotiables include the commitment to these four principles, which must be balanced against one another in cases of conflict. Typical failure modes include the potential for one principle to overshadow others, leading to unjust or harmful outcomes. For example, an excessive focus on efficiency could undermine fairness or autonomy. Critics of Principlism argue that its principles can be too abstract and may lack the specificity needed to guide ethical decision-making effectively.

3) Principle-by-Principle Deep Critical Evaluation

Respect for Autonomy

The use case aligns with the principle of respect for autonomy by advocating for a fair decision-making process that considers the rights of prosumers. By modeling the curtailment decision as a social choice problem, the authors implicitly recognize the need for prosumers to have a voice in how their energy generation is managed. However, the framework could be strengthened by explicitly addressing how informed consent and participation will be facilitated among prosumers. The absence of a clear mechanism for ensuring that all stakeholders can express their preferences may undermine the autonomy of those who are less empowered or informed.

Nonmaleficence

The principle of nonmaleficence is addressed in the use case through the emphasis on minimizing harm to prosumers and ensuring that curtailment practices do not lead to adverse outcomes. The authors argue that existing ad-hoc curtailment rules can result in inefficiencies and disincentivization of investments, which could harm both prosumers and the broader energy system. However, the use case does not sufficiently explore the potential harms that may arise from the proposed curtailment framework itself. For instance, if the curtailment disproportionately affects certain prosumers, this could lead to economic or social harm, which would violate the principle of nonmaleficence.

Beneficence

The principle of beneficence is central to the use case, as it seeks to promote the welfare of prosumers and the community through a fair curtailment mechanism. By deriving curtailment objectives from foundational axioms, the authors aim to ensure that the decision-making process is aligned with the best interests of all stakeholders. However, the framework lacks a detailed exploration of how the proposed social welfare function will effectively promote overall welfare. The authors should provide empirical evidence or theoretical justification for why the KS rule is the most beneficial approach, particularly in terms of its impact on prosumer welfare and grid stability.

Justice

Justice is a key focus of the use case, as it seeks to establish a fair distribution of curtailment burdens among prosumers. The authors argue that existing fairness metrics are insufficient and propose a more rigorous approach grounded in axiomatic principles. While the emphasis on justice is commendable, the framework could benefit from a more nuanced discussion of how different fairness metrics may lead to varying outcomes. For instance, the authors should consider how the choice of fallback and utopia envelopes may affect the distribution of curtailment burdens and whether these choices are themselves justifiable.

4) Stress Tests / Edge Cases

To evaluate the robustness of the proposed curtailment framework, it is essential to consider potential edge cases that may reveal weaknesses or necessitate adjustments. One scenario could involve a situation where a significant number of prosumers are reliant on their PV generation for their livelihood. In this case, any curtailment could have severe economic implications, raising questions about the fairness of the distribution of curtailment burdens. Under Principlism, the framework would need to ensure that the principle of justice is prioritized to protect vulnerable prosumers while balancing the operational constraints of the grid.

Another edge case might involve a situation where the grid experiences unexpected demand spikes, necessitating immediate curtailment decisions. In such a scenario, the principles of nonmaleficence and beneficence would be put to the test, as the need for rapid action could conflict with the desire to ensure fairness and respect for autonomy. The framework would need to provide clear guidelines for how to navigate these conflicts, ensuring that the decision-making process remains transparent and justifiable.

5) Recommendations

In light of the evaluation, several recommendations can be made to enhance the alignment of the proposed curtailment framework with the principles of Principlism. First, the authors should explicitly outline mechanisms for ensuring prosumer participation and informed consent in the decision-making process. This would strengthen the principle of respect for autonomy and ensure that all stakeholders have a voice in how their energy generation is managed.

Second, the framework should include a more comprehensive analysis of the potential harms associated with the proposed curtailment practices. By identifying and addressing these harms, the authors can better align their approach with the principle of nonmaleficence.

Third, the authors should provide empirical evidence or theoretical justification for the effectiveness of the KS rule in promoting overall welfare. This would enhance the alignment with the principle of beneficence and ensure that the proposed framework is grounded in a solid understanding of its impact on prosumer welfare and grid stability.

Finally, the authors should engage in a more nuanced discussion of how different fairness metrics may lead to varying outcomes in terms of justice. By exploring the implications of their choices regarding fallback and utopia envelopes, the authors can better ensure that their approach is justifiable and equitable.

In summary, while the proposed welfarist approach to fair generation curtailment presents a promising framework for addressing the challenges of distributed energy systems, it requires further refinement to fully align with the principles of Principlism. By addressing the identified gaps and enhancing the robustness of their approach, the authors can contribute to a more equitable and sustainable energy future.

Virtue Ethics Framework evaluation

ALIGNMENT	CONCERN	CONFIDENCE	BENCHMARK
55	60	70	**Mixed** The use case demonstrates some alignment with virtue ethics by addressing fairness and social welfare, but it lacks a robust engagement with moral character and virtues, leading to significant concerns about its ethical implications.

Framework Evaluation of A Welfarist Perspective on Fair Generation Curtailment via Virtue Ethics

1) The Use-Case, Reconstructed

The use case presented in "A Welfarist Perspective on Fair Generation Curtailment" addresses the ethical and operational challenges faced by grid operators in managing power curtailment due to the proliferation of distributed photovoltaics (PV). The decision problem revolves around determining which generators to curtail in a manner that is fair, efficient, and aligned with regulatory and social equity considerations. The authors propose a welfarist approach that models this decision as a social choice problem over feasible operating points, aiming to derive curtailment objectives from foundational axioms that express principled stances on fairness and grid access rights.

The target outcomes of this approach include minimizing total curtailment of renewable sources, ensuring fair grid access, promoting social equity, and encouraging investment in renewable generation. Implicit in this framework is the assumption that fairness in curtailment can be quantitatively assessed and that a social welfare function can guide decision-making. The authors argue that existing curtailment schemes often lack a consistent axiomatic foundation, leading to undesirable outcomes such as inefficiency and inequity. By adopting a welfarist stance, they aim to provide a rigorous basis for fair social ranking, thus enhancing transparency and accountability in curtailment decisions.

Key stakeholders in this scenario include grid operators, prosumers (those who both produce and consume energy), and regulatory bodies. The optimization of curtailment decisions privileges the welfare of these stakeholders, particularly focusing on the equitable distribution of grid access rights. The authors suggest that success in this use case is defined by the ability to align curtailment practices with the principles of fairness and efficiency, as determined by the chosen social welfare function.

The use case also reveals hidden assumptions about the comparability of utilities among prosumers. By adopting a cardinal non-comparability stance, the authors imply that while individual utilities can be assessed, direct comparisons across different prosumers may not be valid. This raises questions about how to measure and aggregate these utilities in a way that respects the diverse contexts and circumstances of each prosumer.

2) The Framework, Reconstructed

Virtue ethics, as outlined in the Stanford Encyclopedia of Philosophy, emphasizes the importance of moral character and virtues in ethical decision-making. Unlike consequentialist or deontological frameworks, virtue ethics focuses on what it means to be a good person and how virtues contribute to human flourishing (eudaimonia). The core principles of virtue ethics include the centrality of virtues, the importance of practical wisdom (phronesis), and the foundational role of moral character in ethical considerations.

The framework posits that virtues are excellent traits of character that enable individuals to act in ways that are morally commendable. Practical wisdom is crucial in navigating complex moral situations, as it allows individuals to discern the right course of action based on the specific context. Virtue ethics does not seek to establish rigid rules or outcomes but instead emphasizes the development of moral character and the cultivation of virtues that promote human flourishing.

A critical aspect of virtue ethics is its resistance to defining virtues in terms of other normative concepts, such as consequences or duties. This foundational stance implies that virtues and vices are primary moral categories that inform ethical behavior. The framework also acknowledges the role of emotions and motivations in moral decision-making, emphasizing that virtuous actions arise from a deep understanding of what it means to live well.

However, virtue ethics is not without its challenges. Critics often point to its potential for cultural relativism, as different societies may embody different virtues. Additionally, the framework may struggle to provide clear guidance in situations where virtues conflict, raising questions about how to prioritize competing moral claims. Despite these challenges, virtue ethics remains a compelling approach to ethics, particularly in its focus on character development and the moral dimensions of human relationships.

3) Principle-by-Principle Deep Critical Evaluation

When evaluating the use case through the lens of virtue ethics, several key principles emerge that warrant careful consideration. The first principle is the centrality of virtues in ethical decision-making. The authors of the use case advocate for a welfarist approach that prioritizes utility and fairness, potentially sidelining the importance of moral character and virtues. While the use case emphasizes fairness in curtailment, it does not explicitly engage with the virtues that might underpin such fairness, such as justice, compassion, and integrity.

The second principle is the role of practical wisdom in navigating complex ethical situations. The use case presents a structured framework for decision-making based on social welfare functions, yet it may lack the nuanced understanding that practical wisdom provides. Virtue ethics emphasizes the importance of context and the need for moral agents to exercise discernment in their actions. The reliance on predetermined metrics for fairness may overlook the complexities of individual circumstances and the moral character of the prosumers involved.

Furthermore, the use case implies a quantitative approach to fairness that may not fully capture the qualitative dimensions of moral character. For instance, while the authors argue for a fair distribution of curtailment burdens, they do not address how virtues such as solidarity and community engagement might inform these decisions. The emphasis on utility and efficiency may inadvertently prioritize certain stakeholders over others, potentially undermining the moral fabric of the community.

The third principle is the importance of moral character in ethical considerations. The use case implicitly assumes that fairness can be achieved through mathematical models and social welfare functions, yet it does not account for the moral character of the decision-makers or the prosumers. Virtue ethics emphasizes that ethical behavior arises from the cultivation of virtues, which are deeply rooted in moral character. The decision-making process should not only focus on outcomes but also on the moral development of individuals involved in the curtailment decisions.

Moreover, the use case raises questions about the implications of adopting a cardinal non-comparability stance on utilities. While this approach allows for a more flexible understanding of individual preferences, it may also obscure the moral dimensions of the decisions being made. Virtue ethics encourages a holistic view of moral character that encompasses both individual and communal aspects. The use case's focus on utility may inadvertently promote an individualistic perspective that neglects the relational aspects of ethical decision-making.

In summary, while the use case presents a structured approach to fair generation curtailment, it diverges from the principles of virtue ethics in several key areas. The emphasis on utility and fairness may overshadow the importance of moral character and virtues, while the reliance on predetermined metrics may limit the exercise of practical wisdom. To align more closely with virtue ethics, the use case would benefit from a deeper engagement with the virtues that underpin fairness and a recognition of the moral character of all stakeholders involved.

4) Stress Tests / Edge Cases

To further evaluate the use case against the principles of virtue ethics, it is essential to consider potential stress tests or edge cases that may arise in practice. One scenario involves a situation where a severe weather event leads to an unexpected spike in energy demand. In this case, grid operators may face a dilemma: should they prioritize the needs of prosumers who have invested heavily in renewable energy sources, or should they ensure equitable access for all prosumers, regardless of their investment levels?

From a virtue ethics perspective, this scenario highlights the importance of practical wisdom in navigating competing moral claims. A rigid adherence to predetermined metrics for fairness may lead to decisions that fail to account for the unique circumstances of each prosumer. For instance, prioritizing the needs of those who have made significant investments in renewable energy may be seen as just, but it could also foster resentment among those who feel marginalized by such decisions. A virtue ethics approach would encourage decision-makers to exercise discernment and consider the broader implications of their actions on community relationships and moral character.

Another edge case involves a situation where a prosumer is unable to generate sufficient energy due to unforeseen circumstances, such as equipment failure. In this case, the grid operator must decide whether to curtail the energy of other prosumers to accommodate the needs of the affected individual. A utilitarian approach may prioritize the overall efficiency of the grid, while a virtue ethics perspective would emphasize the importance of compassion and solidarity in supporting the affected prosumer. This scenario underscores the need for a more nuanced understanding of fairness that incorporates the moral character of decision-makers and the relational dynamics of the community.

These stress tests reveal that the use case may benefit from a more flexible approach that allows for the exercise of practical wisdom and moral character in decision-making. By engaging with the virtues that underpin fairness and considering the unique circumstances of each prosumer, grid operators can foster a more equitable and just energy system.

5) Recommendations

In light of the evaluation, several recommendations emerge to enhance the alignment of the use case with the principles of virtue ethics. First, the authors should explicitly incorporate a discussion of the virtues that underpin fairness in curtailment decisions. This could involve identifying key virtues such as justice, compassion, and solidarity, and exploring how these virtues can inform decision-making processes. By grounding the use case in a robust understanding of moral character, the authors can create a more holistic framework for fair generation curtailment.

Second, the use case would benefit from a greater emphasis on practical wisdom in navigating complex ethical situations. This could involve developing guidelines for decision-makers that encourage discernment and reflection on the moral implications of their actions. By fostering an environment that values practical wisdom, grid operators can better address the unique circumstances of each prosumer and promote a more equitable distribution of curtailment burdens.

Third, the authors should consider incorporating qualitative measures of fairness alongside quantitative metrics. While the use of social welfare functions provides a structured approach to decision-making, it may overlook the moral dimensions of individual circumstances. By integrating qualitative assessments of fairness, the authors can create a more nuanced understanding of the ethical implications of curtailment decisions.

Lastly, the use case should explicitly address the moral character of decision-makers and the importance of fostering a culture of ethical responsibility within the grid operator's organization. This could involve training programs that emphasize the cultivation of virtues and the development of moral character among decision-makers. By prioritizing ethical leadership, grid operators can ensure that fairness in curtailment decisions is grounded in a commitment to moral integrity and social responsibility.

In conclusion, while the use case presents a compelling approach to fair generation curtailment, it diverges from the principles of virtue ethics in several key areas. By incorporating a deeper engagement with virtues, emphasizing practical wisdom, integrating qualitative measures of fairness, and fostering ethical leadership, the authors can create a more robust framework that aligns with the core commitments of virtue ethics.

Care Ethics Framework evaluation

ALIGNMENT	CONCERN	CONFIDENCE	BENCHMARK
55	60	70	**Mixed** The use case demonstrates a commitment to fairness and social equity but lacks depth in addressing individual needs and perspectives, leading to significant concerns about its practical application and potential inequities.

Framework Evaluation of A Welfarist Perspective on Fair Generation Curtailment via Care Ethics

1) The Use-Case, Reconstructed

The use case presented in "A Welfarist Perspective on Fair Generation Curtailment" addresses the ethical and operational challenges associated with power curtailment in distribution grids that incorporate distributed photovoltaics (PV). The decision problem revolves around how to fairly allocate the curtailment of power generation among prosumers—individuals or entities that both produce and consume energy—while adhering to grid operational constraints. The authors propose a welfarist approach to curtailment, which is modeled as a social choice problem over feasible operating points. The aim is to derive curtailment objectives from foundational axioms that express principled stances on fairness and grid access rights.

The target outcomes of this approach include the minimization of total curtailment, equitable access to the grid, and the promotion of social equity among prosumers. The authors argue that existing curtailment rules often lack a consistent axiomatic foundation, leading to inefficiencies and inequities. By adopting a cardinal non-comparability stance on utilities, the authors seek to reduce the assumptions about prosumers' private preferences while providing a rigorous basis for fair social ranking.

Implicit value claims in the use case include the prioritization of fairness and social equity over purely economic considerations. The authors suggest that fairness in curtailment should not merely be about maximizing efficiency but should also account for the welfare of all stakeholders involved. The decision context involves various stakeholders, including grid operators, prosumers, and regulatory bodies, each with distinct interests and stakes in the outcome. The use case assumes that success is defined not only by operational efficiency but also by the equitable treatment of all prosumers, particularly those who may be more vulnerable or dependent on the grid.

The authors also imply that the existing curtailment schemes are often ad-hoc and lack transparency, which can lead to undesirable outcomes such as disincentivizing future investments in renewable energy generation. The proposed welfarist framework aims to rectify these issues by providing a structured, auditable, and principled approach to curtailment decisions.

2) The Framework, Reconstructed

Care ethics, as articulated in the provided framework, emphasizes the moral significance of relationships and dependencies in human life. It posits that ethical considerations should focus on maintaining relationships and promoting the well-being of both caregivers and care-receivers within a network of social relations. Care ethics is often framed as a practice or virtue rather than a strict moral theory, highlighting the importance of emotional engagement, contextual reasoning, and the particularities of individual situations.

The core principles of care ethics include attentiveness to the needs of others, the responsibility to respond to those needs, competence in providing care, and responsiveness to the perspectives of those being cared for. This framework prioritizes relational dynamics over abstract principles, suggesting that moral reasoning should be grounded in the lived experiences of individuals and the specific contexts in which they operate.

Care ethics also critiques traditional moral theories, such as Kantian deontology and utilitarianism, for their potential biases and limitations. It highlights the importance of considering the voices of those who are often marginalized in ethical discussions, particularly women and other vulnerable groups. Critics of care ethics have raised concerns about its potential essentialism, parochialism, and ambiguity, suggesting that it may inadvertently reinforce traditional gender roles or overlook broader social dynamics.

The framework's non-negotiables include a commitment to relationality, context, and the importance of emotional engagement in moral deliberation. It emphasizes that ethical considerations should not be abstracted from the realities of human relationships and dependencies. Care ethics also recognizes the need for justice and equity, particularly in addressing the needs of the most vulnerable members of society.

3) Principle-by-Principle Deep Critical Evaluation

In evaluating the use case through the lens of care ethics, several key principles emerge that warrant careful consideration. The first principle is attentiveness to the needs of prosumers. The use case emphasizes the importance of fairness in curtailment decisions, which aligns with the care ethics principle of being attentive to the needs of individuals. However, the framework requires a deeper exploration of how the specific needs of different prosumers are identified and prioritized. The authors assert that fairness should be a guiding principle, but they do not sufficiently elaborate on how to operationalize this attentiveness in practice. For instance, how do we determine which prosumers are most vulnerable or dependent on the grid? Without clear criteria for assessing needs, the proposed framework risks being overly abstract and may fail to adequately address the complexities of individual circumstances.

The second principle is the responsibility to respond to the needs of others. The authors argue for a welfarist approach that seeks to minimize total curtailment while ensuring equitable access to the grid. This aligns with the care ethics emphasis on mutual responsibility in relationships. However, the use case does not adequately address the potential conflicts that may arise between the needs of different prosumers. For example, if one prosumer's need for energy is prioritized over another's, how is that decision justified within the framework of care ethics? The authors need to provide a more nuanced discussion of how to balance competing needs and responsibilities among prosumers.

The third principle is competence in providing care. The authors propose a structured, axiomatic approach to curtailment decisions, which suggests a level of competence in addressing the complexities of the energy grid. However, the framework does not sufficiently engage with the practical implications of implementing these decisions. For instance, what training or resources are required for grid operators to effectively apply the proposed welfarist framework? Additionally, the authors should consider the potential for unintended consequences that may arise from their proposed solutions. Care ethics emphasizes the importance of being aware of the broader social dynamics at play, and the authors should critically examine how their approach may inadvertently reinforce existing power imbalances or inequities.

The fourth principle is responsiveness to the perspectives of those being cared for. The authors advocate for a fair social ranking of curtailment decisions, which aligns with the care ethics emphasis on considering the voices of all stakeholders. However, the use case lacks a clear mechanism for incorporating the perspectives of prosumers into the decision-making process. How can prosumers actively participate in shaping the curtailment criteria that affect their lives? The authors should explore ways to engage prosumers in meaningful dialogue and decision-making, ensuring that their voices are heard and valued.

In summary, while the use case demonstrates a commitment to fairness and social equity, it falls short in several key areas when evaluated through the lens of care ethics. The framework demands a more nuanced understanding of the complexities of individual needs, responsibilities, and perspectives. By addressing these gaps, the authors can strengthen their alignment with care ethics and enhance the ethical robustness of their proposed solutions.

4) Stress Tests / Edge Cases

To further evaluate the use case under the framework of care ethics, it is essential to consider potential stress tests or edge cases that may reveal weaknesses or necessitate adjustments to the proposed approach.

One relevant scenario involves a situation where a severe weather event leads to a sudden increase in energy demand while simultaneously causing disruptions to the grid. In this case, the grid operator must make rapid decisions about curtailment. The use case suggests a structured approach to curtailment based on a welfarist framework, but it does not account for the urgency and unpredictability of such situations. Care ethics emphasizes the importance of responsiveness and attentiveness to immediate needs, which may conflict with the more deliberative approach proposed in the use case. If the grid operator prioritizes efficiency over the immediate needs of vulnerable prosumers during a crisis, the ethical implications could be significant. This scenario highlights the need for flexibility and adaptability in the proposed framework, allowing for rapid responses that prioritize the well-being of those most affected by the crisis.

Another edge case involves the potential for inequitable outcomes based on socio-economic disparities among prosumers. For instance, if wealthier prosumers are able to invest in larger PV systems, they may be less affected by curtailment decisions compared to lower-income prosumers who rely more heavily on the grid. The use case advocates for fairness in curtailment, but it does not adequately address the systemic inequalities that may influence the distribution of benefits and burdens among prosumers. Care ethics demands a critical examination of how power dynamics and socio-economic factors shape the experiences of individuals within the grid system. This scenario underscores the importance of incorporating a justice-oriented perspective into the proposed framework, ensuring that the needs of the most vulnerable are prioritized and that inequities are actively addressed.

In both scenarios, the use case reveals potential weaknesses in its alignment with care ethics. The framework requires a more nuanced understanding of the complexities of human relationships and dependencies, particularly in times of crisis or systemic inequality. By addressing these edge cases, the authors can enhance the ethical robustness of their proposed solutions and ensure that they are responsive to the diverse needs of all prosumers.

5) Recommendations

In light of the evaluation, several recommendations can be made to enhance the alignment of the use case with the principles of care ethics.

First, the authors should incorporate a more explicit framework for assessing the needs of prosumers. This could involve developing criteria for identifying vulnerable or dependent individuals within the grid system. By establishing clear metrics for assessing needs, the authors can ensure that their approach is grounded in the realities of individual circumstances and relationships.

Second, the use case should address the potential conflicts that may arise between the needs of different prosumers. The authors could explore mechanisms for balancing competing needs and responsibilities, such as establishing a decision-making framework that prioritizes the most vulnerable while still considering the broader context of grid operations.

Third, the authors should engage with prosumers in meaningful dialogue and decision-making processes. This could involve creating platforms for prosumers to voice their concerns and preferences regarding curtailment decisions. By actively involving prosumers in the decision-making process, the authors can ensure that their perspectives are valued and that the proposed solutions are more responsive to their needs.

Fourth, the authors should consider the implications of urgent situations, such as natural disasters, on their proposed framework. Developing guidelines for rapid responses that prioritize the well-being of vulnerable prosumers in crisis situations would enhance the ethical robustness of their approach.

Finally, the authors should critically examine the potential for systemic inequalities to influence curtailment outcomes. By incorporating a justice-oriented perspective into their framework, the authors can ensure that their approach actively addresses inequities and promotes social equity among prosumers.

By implementing these recommendations, the authors can strengthen the alignment of their proposed solutions with the principles of care ethics, enhancing the ethical robustness of their approach to fair generation curtailment.

Communitarian Ethics

Framework evaluation

ALIGNMENT	CONCERN	CONFIDENCE	BENCHMARK
65	45	70	**Moderate Alignment** The use case aligns moderately with communitarian ethics by emphasizing community engagement and fairness. However, it diverges in its reliance on a welfarist perspective that may overlook the diversity of community values and the importance of participatory decision-making.

Framework Evaluation of A Welfarist Perspective on Fair Generation Curtailment via Communitarian Ethics

1) The Use-Case, Reconstructed

The use case presented, titled "A Welfarist Perspective on Fair Generation Curtailment," addresses the decision-making challenges faced by grid operators in managing distributed photovoltaics (PV) within local energy systems. The primary decision problem revolves around how to implement active power curtailment in a manner that is perceived as fair among the various prosumers—individuals or entities that both produce and consume electricity. The proposed mechanism involves modeling the curtailment decision as a social choice problem, where the objective is to derive curtailment objectives from foundational axioms that express fairness and grid access rights.

The target outcomes of this approach are twofold: first, to ensure that curtailment decisions are made transparently and justifiably; and second, to promote social equity among prosumers by minimizing the total curtailment of renewable sources. The authors argue that existing ad-hoc curtailment rules lack a consistent axiomatic foundation, leading to inefficiencies and inequities. By adopting a welfarist stance, the authors aim to create a social welfare function that can guide the decision-making process in a principled manner.

Implicit value claims in the use case include the belief that fairness is essential for social cohesion and that equitable access to energy resources is a moral imperative. The authors assume that prosumers have a right to fair treatment in curtailment decisions, which implies a commitment to social equity. The optimization of fairness and grid access rights is framed as a moral obligation, suggesting that success is defined not merely by technical efficiency but by the perceived justice of the outcomes.

Stakeholders in this scenario include grid operators, prosumers, and potentially regulatory bodies that govern energy distribution. The optimization process is framed around the idea that fairness should be prioritized alongside technical efficiency, with the understanding that social equity can lead to greater investment in renewable energy sources. The hidden assumptions in this use case include the belief that all prosumers have equal stakes in the energy system and that their utilities can be compared in a meaningful way without extensive calibration.

2) The Framework, Reconstructed

Communitarian ethics posits that human identities are largely shaped by the communities and social relations in which individuals are embedded. This framework emphasizes the importance of community in moral and political judgments, arguing that individuals derive meaning and ethical guidance from their social contexts. The core principles of communitarian ethics include the recognition of the social nature of the self, the value of particular communities, and the necessity of nurturing these communities to foster moral and political engagement.

One of the framework's non-negotiables is the rejection of strict individualism, which is often associated with liberal theories. Communitarianism asserts that moral and political reasoning must be informed by the traditions, histories, and values of specific communities. This perspective challenges the notion that universal principles can adequately address the complexities of human relationships and social obligations.

The framework treats community as morally primary, emphasizing that individuals are not isolated entities but are deeply interconnected through various social ties. This interconnectedness informs their moral responsibilities and obligations. Critics of communitarianism often argue that it can lead to parochialism or the suppression of individual rights in favor of collective norms. However, communitarian thinkers contend that a balance can be struck between individual autonomy and communal obligations.

In terms of decision procedures, communitarian ethics advocates for deliberative processes that engage community members in discussions about moral and political issues. This participatory approach is seen as essential for fostering a sense of belonging and shared responsibility. The framework's limits include the challenge of reconciling diverse community values and the potential for conflict between communal norms and individual rights.

3) Principle-by-Principle Deep Critical Evaluation

The use case aligns with several core principles of communitarian ethics, particularly in its emphasis on community and social relations. By framing the curtailment decision as a social choice problem, the authors acknowledge the importance of collective decision-making and the need for transparency in the process. This aligns with the communitarian view that moral and political judgments should be informed by the values and traditions of specific communities.

However, the use case diverges from communitarian principles in its reliance on a welfarist perspective, which emphasizes individual utilities and social welfare functions. While the authors argue for a fair distribution of curtailment burdens, the underlying assumption that utilities can be compared across prosumers may overlook the rich tapestry of community values that inform individual preferences. Communitarian ethics would demand a more nuanced understanding of how different communities perceive fairness and equity, rather than relying solely on a utility-based framework.

The use case also implies a level of homogeneity among prosumers that may not exist in practice. Communitarian ethics would challenge this assumption by highlighting the diverse values and priorities that different communities may hold regarding energy production and consumption. The authors' focus on minimizing total curtailment may not adequately account for the specific needs and historical contexts of various prosumer communities.

Moreover, the use case's emphasis on technical efficiency may conflict with the communitarian principle that prioritizes communal well-being over individual gain. While the authors advocate for a fair distribution of curtailment burdens, the framework's reliance on a social welfare function may inadvertently prioritize efficiency over the moral obligations that arise from community ties. This tension raises questions about whether the proposed mechanism can truly embody the spirit of communitarian ethics.

In terms of specific behaviors, the use case suggests that grid operators should adopt a social welfare function that minimizes total curtailment. However, this approach may not adequately reflect the diverse perspectives within communities. Communitarian ethics would advocate for a more participatory approach, where prosumers are actively engaged in discussions about curtailment decisions and their implications for community life.

The framework also demands that the decision-making process be transparent and accountable to community members. While the use case acknowledges the importance of transparency, it does not provide sufficient detail on how prosumers will be involved in the decision-making process. This lack of engagement could undermine the legitimacy of the curtailment decisions and perpetuate feelings of disenfranchisement among prosumers.

4) Stress Tests / Edge Cases

To further evaluate the robustness of the use case under communitarian ethics, two relevant scenarios can be conceived. The first scenario involves a community with a strong historical commitment to renewable energy and a collective identity centered around sustainability. In this context, prosumers may prioritize local energy production and resist curtailment measures that threaten their communal values. If the proposed curtailment mechanism fails to account for these deeply held beliefs, it may lead to significant pushback from the community, undermining the legitimacy of the grid operator's decisions.

In this scenario, the communitarian framework would emphasize the need for the grid operator to engage with the community in a meaningful way, fostering dialogue and understanding. The failure to do so could reveal a principled flaw in the use case, as it would neglect the moral obligations that arise from community ties and shared values.

The second scenario involves a community with diverse energy needs and varying levels of investment in renewable technologies. In this case, the use case's reliance on a social welfare function may lead to outcomes that disproportionately benefit certain prosumers while disadvantaging others. For example, if the curtailment decisions prioritize efficiency over equity, some prosumers may bear a heavier burden than others, leading to tensions within the community.

Under communitarian ethics, this scenario would highlight the importance of recognizing the unique needs and values of different community members. The failure to address these disparities could result in a breakdown of social cohesion and a sense of injustice among prosumers. The communitarian framework would demand a more nuanced approach that considers the diverse perspectives within the community and seeks to promote fairness in a way that aligns with communal values.

5) Recommendations

In light of the evaluation, several recommendations can be made to enhance the alignment of the use case with communitarian ethics. First, the authors should consider incorporating a more participatory decision-making process that actively engages prosumers in discussions about curtailment measures. This could involve community forums or workshops where prosumers can voice their concerns and preferences, fostering a sense of ownership over the decision-making process.

Second, the use case should acknowledge the diversity of values and priorities within prosumer communities. By recognizing that different communities may have distinct perspectives on fairness and equity, the authors can develop a more nuanced approach that respects these differences. This could involve tailoring curtailment measures to reflect the specific needs and historical contexts of various communities.

Third, the authors should consider revising the social welfare function to incorporate communal values and priorities. Rather than relying solely on individual utilities, the framework could be expanded to include metrics that reflect the collective well-being of the community. This would align the decision-making process with the communitarian principle that prioritizes communal obligations over individual gain.

Finally, the use case should emphasize the importance of transparency and accountability in the decision-making process. By providing clear explanations of how curtailment decisions are made and the rationale behind them, grid operators can foster trust and legitimacy within the community. This transparency is essential for ensuring that prosumers feel heard and valued in the decision-making process.

In conclusion, while the use case presents a compelling approach to fair generation curtailment, it requires significant revisions to align with the principles of communitarian ethics. By prioritizing community engagement, recognizing diversity, and incorporating communal values into the decision-making process, the authors can create a more just and equitable framework for managing distributed photovoltaics in local energy systems.

Casuistry Framework evaluation

ALIGNMENT	CONCERN	CONFIDENCE	BENCHMARK
65	45	70	**Moderate Alignment** The use-case demonstrates a moderate alignment with the principles of casuistry, particularly in its emphasis on case-based reasoning and moral intuition. However, concerns regarding interpersonal comparability and the moral agent's role indicate significant unresolved tensions.

Framework Evaluation of A Welfarist Perspective on Fair Generation Curtailment via Casuistry

1) The Use-Case, Reconstructed

The use-case presented is a welfarist approach to fair active power curtailment in distribution grids with distributed photovoltaics (PV). The decision problem at hand involves determining which generators should be curtailed to respect grid operational constraints, particularly in scenarios where the injection of active power exceeds the grid's capacity. The authors propose a mechanism that models this decision as a social choice problem, deriving curtailment objectives from foundational axioms that express principled stances on fairness and grid access rights.

The target outcomes of this approach include achieving a fair distribution of curtailment among prosumers, minimizing total curtailment of renewable sources, and ensuring non-discriminatory access to the grid. The authors argue that existing curtailment rules often lack consistent axiomatic foundations, leading to undesirable outcomes such as inefficiency and disincentivization of future investments. Thus, the proposed solution aims to provide a rigorous basis for fair social ranking while addressing the complexities of heterogeneous residential systems.

Implicit value claims in the use-case include the prioritization of fairness, equity, and sustainability in energy distribution. The authors assume that fairness is not merely a matter of equal treatment but must also account for the varying capacities and contributions of different prosumers. The optimization of curtailment decisions is framed within a welfarist perspective, suggesting that the welfare of prosumers should guide the decision-making process.

The stakeholders involved in this decision-making process include grid operators, prosumers (households or entities that both produce and consume energy), and regulatory bodies. The success of the proposed mechanism is measured by its ability to balance the competing criteria of sustainability, fair grid access, social equity, and investment promotion. However, underlying assumptions include the comparability of utilities among prosumers, which may not hold in practice, particularly in heterogeneous systems.

The use-case also implies that the decision-making process can be streamlined through the adoption of a social welfare function (SWF) that reflects the welfare of all agents involved. This SWF is derived from a set of axioms that guide the selection of curtailment schemes, thereby providing a structured approach to what is often an ad-hoc decision-making process.

2) The Framework, Reconstructed

The framework of casuistry, as articulated in the provided overview, is a method of ethical reasoning that emphasizes case-based analysis and moral intuition. It seeks to derive generalizable principles from specific cases rather than applying universal maxims derived from high-level ethical theories. Casuistry is particularly relevant in contexts where moral pluralism exists, allowing for the consideration of diverse ethical perspectives without being constrained by a single theoretical framework.

Core principles of casuistry include the analysis of moral issues through paradigms and analogies, leading to the formulation of expert opinions about moral obligations. The methodology involves a three-step process: first, a detailed description of the moral dilemma; second, categorization and comparison of the case with past paradigms; and third, reaching a moral judgment through inductive reasoning and moral intuition.

The framework prioritizes moral intuition and the particulars of a case over abstract principles, allowing for flexibility and adaptability in ethical decision-making. This is particularly important in the context of globalization, where diverse cultural and ethical perspectives must be considered. However, this flexibility can also lead to weaknesses, such as the potential for moral emotivism and the exclusion of the moral agent's internal motivations and intentions.

Non-negotiables within this framework include the rejection of high-level ethical theories that impose rigid structures on moral reasoning. Casuistry emphasizes the importance of context and the specifics of each case, which can lead to a more nuanced understanding of ethical dilemmas. However, this approach also faces criticism for potentially neglecting the moral agent's role and the internal aspects of moral decisions.

Typical objections to casuistry include concerns about its lack of a clear theoretical foundation, which can lead to inconsistencies in moral judgments. Critics argue that without a robust ethical theory to guide decision-making, casuistry may devolve into subjective interpretations of fairness and morality. Additionally, the framework's reliance on moral intuition raises questions about the objectivity and universality of its conclusions.

3) Principle-by-Principle Deep Critical Evaluation

The welfarist approach to fair generation curtailment aligns with several principles of casuistry, particularly in its emphasis on case-based reasoning and the importance of context. The use-case seeks to address the complexities of curtailment decisions by modeling them as social choice problems, which resonates with the casuistic focus on analyzing specific cases rather than applying universal principles.

One area of alignment is the emphasis on moral intuition. The authors argue for a social welfare function that reflects the welfare of all prosumers, which can be seen as an appeal to moral intuition about fairness and equity. This aligns with the casuistic principle of deriving moral judgments from the particulars of a case, as the proposed mechanism seeks to account for the diverse circumstances of different prosumers.

However, there are notable divergences between the use-case and the principles of casuistry. One significant divergence lies in the assumption of interpersonal comparability of utilities among prosumers. The welfarist approach relies on the ability to compare utilities across different agents, which may not hold true in heterogeneous systems where individual preferences and values vary significantly. This assumption could lead to ethical tensions, as it may overlook the unique circumstances and moral intuitions of individual prosumers.

Furthermore, the use-case's reliance on a social welfare function may inadvertently impose a form of abstraction that casuistry seeks to avoid. While the authors aim to provide a rigorous basis for fair social ranking, the focus on a singular SWF could limit the flexibility that casuistry promotes. The framework's strength lies in its adaptability to diverse moral intuitions, and the use-case's emphasis on a specific metric may constrain this adaptability.

Another critical point of evaluation is the potential for moral emotivism within the proposed mechanism. The authors argue for a fair distribution of curtailment based on welfare considerations, but without a robust theoretical foundation, the decision-making process could devolve into subjective interpretations of fairness. This concern aligns with the criticisms of casuistry, which may struggle to provide consistent moral guidance in the absence of clear ethical principles.

The use-case also raises questions about the moral agent's role in the decision-making process. While the authors emphasize the importance of fairness and equity, the framework of casuistry highlights the internal aspects of moral decisions, including the intentions and motivations of the moral agents involved. By focusing primarily on external outcomes, the proposed mechanism may overlook the complexities of moral agency and the development of virtues and vices through repeated ethical decisions.

In terms of success criteria, the use-case defines success as achieving a fair distribution of curtailment while minimizing total curtailment of renewable sources. However, this metric may not fully capture the moral dimensions of the decision-making process. Success should also consider the implications of curtailment decisions on the moral development of prosumers and the broader societal values at play.

To address these divergences, the use-case could benefit from a more nuanced understanding of interpersonal comparability and the moral agent's role. Incorporating a broader range of moral intuitions and perspectives could enhance the alignment with casuistry's principles. Additionally, a more explicit acknowledgment of the limitations of the proposed social welfare function could help mitigate concerns about moral emotivism and the potential for subjective interpretations of fairness.

4) Stress Tests / Edge Cases

To further evaluate the robustness of the proposed mechanism, two relevant scenarios can be conceived that may reveal potential flaws or necessitate adjustments to the use-case.

The first scenario involves a situation where a significant disparity exists among prosumers in terms of their energy generation capacity and reliance on the grid. For instance, consider a community where some prosumers have invested heavily in solar panels and are able to generate excess energy, while others rely solely on grid electricity. In this case, the welfarist approach may struggle to achieve a fair distribution of curtailment, as the social welfare function may disproportionately favor those with higher generation capacities. This could lead to ethical tensions, as prosumers with lower capacities may feel marginalized or unfairly treated. Under the framework of casuistry, the decision-making process should account for the unique circumstances of each prosumer, emphasizing the importance of context and moral intuition. The proposed mechanism may need to be adjusted to ensure that it does not inadvertently reinforce existing inequalities within the community.

The second scenario involves a sudden increase in energy demand due to external factors, such as a heatwave or a public event. In this case, the grid may face unprecedented strain, necessitating immediate curtailment decisions. The welfarist approach may struggle to adapt quickly to these changing circumstances, as the reliance on a predetermined social welfare function could hinder responsiveness. Casuistry's emphasis on case-based reasoning and moral intuition may provide a more flexible framework for navigating such dynamic situations. The proposed mechanism could benefit from incorporating a more adaptive decision-making process that allows for real-time adjustments based on the evolving context.

Both scenarios highlight the importance of considering the complexities of moral agency and the need for a more nuanced understanding of fairness in the decision-making process. The proposed mechanism should be refined to ensure that it remains responsive to the diverse needs and circumstances of prosumers, particularly in situations where external factors may significantly impact energy distribution.

5) Recommendations

In light of the evaluation, several recommendations can be made to enhance the alignment of the proposed mechanism with the principles of casuistry.

First, the use-case should explicitly acknowledge the limitations of interpersonal comparability among prosumers. By recognizing the unique circumstances and moral intuitions of individual agents, the decision-making process can be better aligned with casuistry's emphasis on context and case-based reasoning. This could involve incorporating qualitative assessments of prosumers' situations alongside quantitative metrics, allowing for a more holistic understanding of fairness.

Second, the reliance on a singular social welfare function should be reconsidered. While the authors aim to provide a rigorous basis for fair social ranking, the focus on a specific metric may constrain the flexibility that casuistry promotes. Instead, the proposed mechanism could benefit from a multi-faceted approach that allows for the integration of diverse moral intuitions and perspectives. This would enhance the adaptability of the decision-making process and better reflect the complexities of the ethical landscape.

Third, the moral agent's role should be more explicitly integrated into the decision-making process. By considering the internal motivations and intentions of prosumers, the proposed mechanism can better account for the development of virtues and vices through repeated ethical decisions. This could involve incorporating feedback mechanisms that allow prosumers to express their values and preferences, fostering a more participatory approach to decision-making.

Finally, the proposed mechanism should remain responsive to changing circumstances and external factors. Incorporating adaptive decision-making processes that allow for real-time adjustments based on evolving contexts would enhance the mechanism's robustness and responsiveness. This aligns with casuistry's emphasis on flexibility and the importance of context in ethical decision-making.

By implementing these recommendations, the proposed mechanism can better align with the principles of casuistry, enhancing its effectiveness in addressing the complexities of fair generation curtailment in distribution grids with distributed photovoltaics.